

Survey paper

A survey on explainable AI for semantic communication: Architecture, challenges, and future opportunities

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ABSTRACT

As communication systems evolve toward 6G, semantic communication is emerging as a transformative paradigm that prioritizes the accurate transmission of meaning rather than just bits. While artificial intelligence enables this shift by facilitating intelligent interpretation and context-aware processing, it also introduces significant challenges related to transparency, reliability, and user trust. XAI has thus become essential in making AI-enabled semantic communication systems more interpretable, auditable, and accountable. To the best of our knowledge, this is the first survey that systematically analyzes how explainability can be embedded across all stages of the semantic communication pipeline, integrating architectural design, metrics, security considerations, and human-in-the-loop mechanisms. Additionally, the survey identifies pressing research challenges, including the lack of standardization, real-time applicability, and vulnerabilities introduced by opaque AI models. By drawing attention to these issues and outlining future research directions, this survey aims to guide the development of responsible and trustworthy semantic communication systems for next-generation wireless networks.

1. Introduction

Semantic communication (SemCom) introduces a fundamental rethinking of how information exchange is designed and evaluated, departing from classical communication paradigms that primarily focus on bit-level fidelity and raw data rates. Rather than treating information as a sequence of bits alone, semantic communication seeks to transmit the underlying meaning, intent, and contextual significance of messages, thereby redefining the notion of communication efficiency [1]. By prioritizing semantics over syntax, this paradigm enables a more purposeful use of communication resources, reduces unnecessary redundancy, and increases the practical value of transmitted information. As contemporary digital services increasingly depend on accurate contextual awareness and rapid interpretation, semantic communication has become a key enabler for a wide range of emerging applications, including autonomous driving, augmented and virtual reality, remote medical services, smart city infrastructures, and large-scale Internet of Things (IoT) ecosystems [2]. In such application domains, reliable semantic exchange is not merely advantageous; it directly impacts system functionality, operational safety, and overall user experience [3,4].

The incorporation of artificial intelligence into semantic communication architectures further extends their potential by enabling advanced semantic representation learning, adaptive compression, and context-aware inference. Through deep learning and natural language processing techniques, AI-driven SemCom systems learn to interpret, encode, and reconstruct information in a manner that reflects its semantic importance rather than its raw structure. This capability allows communication processes to become more resilient and adaptive, particularly in complex and dynamically changing environments. At the same time, the growing reliance on highly complex AI models introduces pressing concerns related to transparency, interpretability, and trust. Many state-of-the-art models operate as so-called “black-boxes,” making it difficult to understand how specific decisions are produced or to verify their correctness. Such opacity complicates debugging, limits accountability, and can erode user confidence, especially in safety-critical scenarios where erroneous decisions may have serious consequences [5,6].

Mitigating these challenges requires mechanisms that can expose and clarify the internal reasoning of AI-enabled semantic communication systems. Explainable artificial intelligence (XAI) addresses this need by providing methods that translate complex model behavior into

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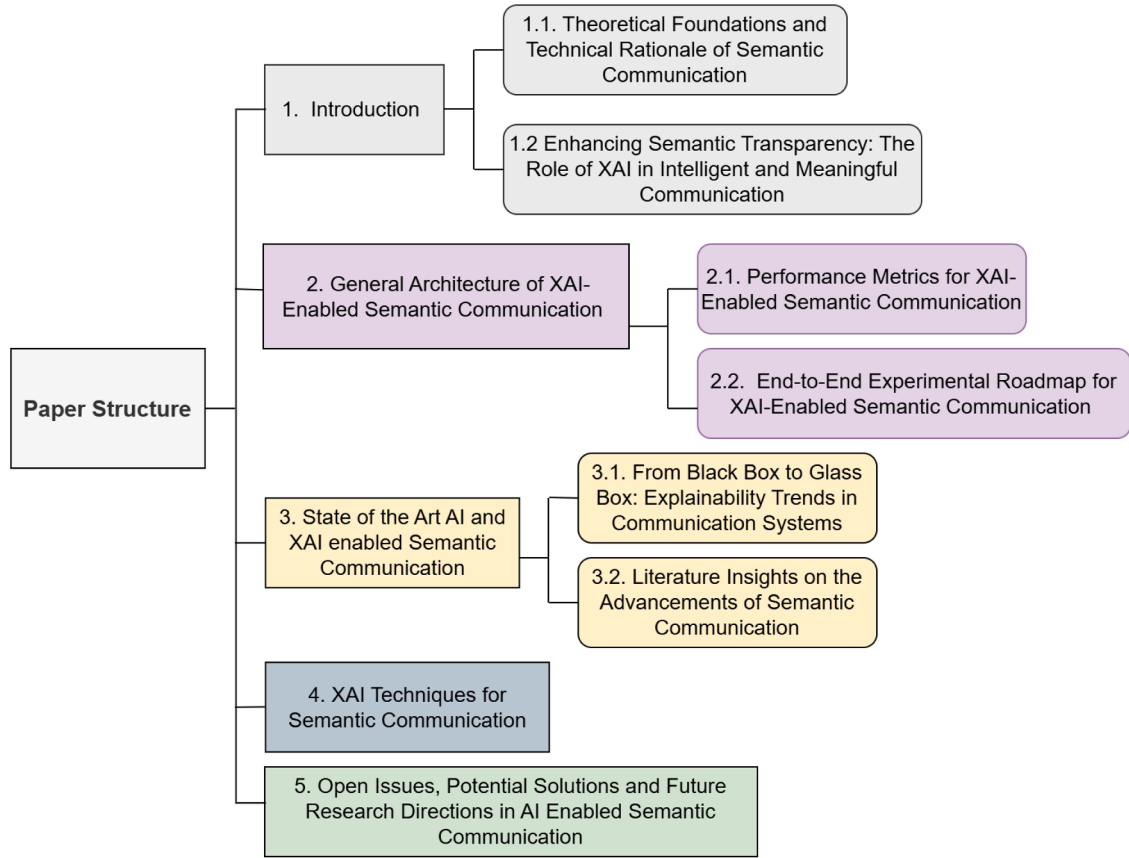


Fig. 1. Paper structure.

Table 1

Mapping the contributions of this survey against gaps identified in prior surveys on semantic communication and explainable AI (XAI).

Contribution of This Survey	Gap Observed in Prior Literature	Representative Prior Works	How This Survey Complements Existing Work
1) Formal framing of explainability in SemCom	Prior surveys typically treat SemCom and XAI as separate topics, without explicitly framing explainability as a first-class objective within semantic communication.	SemCom surveys: [10–13]; XAI surveys: [14–17]	Provides a unified framing that embeds explainability alongside semantic fidelity and system constraints, clarifying what should be explained in SemCom and why fidelity–explainability trade-offs arise.
2) Taxonomy linking XAI methods to SemCom modules	Existing surveys list XAI techniques and SemCom blocks separately, offering limited guidance on how XAI aligns with different pipeline stages (encoder, latent space, channel reasoning, decoder, adaptation).	SemCom: [18–21]; XAI: [16,22,23]	Introduces a cross-layer taxonomy mapping XAI families to SemCom functional modules, enabling more systematic integration decisions (what to explain, where, and with which method).
3) XAI-integrated SemCom architecture	Prior architectural overviews emphasize semantic encoding/decoding, with limited clarity on explanation generation points and how explanations propagate and support decisions.	[24–27]	Presents an end-to-end architecture that explicitly incorporates explanation generation and consumption within the SemCom pipeline, improving transparency and traceability across modules.
4) Evaluation metrics for semantic fidelity and explainability	Prior surveys summarize semantic similarity or task-level metrics, but rarely connect them to explainability quality (faithfulness, stability, compactness, calibration) in SemCom settings.	[13,28]	Consolidates SemCom evaluation metrics and aligns them with explainability-oriented criteria to support transparent and reproducible assessment of XAI-enabled SemCom systems.
5) Security and privacy implications of explainable SemCom	SemCom security and XAI security often evolve separately; explanation leakage and manipulation risks are underrepresented in SemCom survey literature.	[29–31]	Highlights explanation-induced attack surfaces (information leakage, attribution manipulation) and discusses mitigation directions consistent with secure communication principles.
6) Human-in-the-loop feedback and adaptation under explainability	Feedback/adaptation mechanisms are discussed in SemCom, but the role of explanations in shaping user trust, corrections, and stable adaptation is insufficiently examined.	[32–35]	Analyzes how explanations influence user feedback and adaptive updates, highlighting stability considerations and design criteria for trustworthy human-in-the-loop SemCom.
7) Positioning within emerging 6G and MANO research	XAI is increasingly studied for 6G management/orchestration, yet semantic-layer explainability for communication tasks is not consistently linked to these trends.	[36–39]	Connects explainable SemCom to broader 6G intelligence and MANO directions, motivating semantic-layer transparency for trustworthy automation.
8) Illustrative discussion of explainability under channel variation	Many surveys remain conceptual and provide limited intuition on how explanation behavior changes under channel dynamics while semantic outputs may remain stable.	[40–44]	Provides illustrative discussion to ground the SemCom–XAI linkage, helping readers interpret explanation behavior in dynamic wireless environments.

Table 2
List of common acronyms.

Acronym	Description	Acronym	Description
AI	Artificial Intelligence	MAML	Model-Agnostic Meta-Learning
ANN	Artificial Neural Network	MADRL	Multi-Agent Deep Reinforcement Learning
AR	Augmented Reality	MAP	Mean Average Precision
AWGN	Additive White Gaussian Noise	MetaReg	Meta Regularization
BERT	Bidirectional Encoder Representations from Transformers	MEC	Multi-access Edge Computing
BER	Bit Error Rate	MIG	Mutual Information Gap
BLEU	Bilingual Evaluation Understudy Score	MI	Mutual Information
CAV	Concept Activation Vector	MIMO	Multiple Input Multiple Output
CNN	Convolutional Neural Network	MLDG	Model-Agnostic Meta-Learning for Domain Generalization
COPA	Contrastive Prototype Alignment	ML	Machine Learning
CV	Computer Vision	MLOps	Machine Learning Operations
DAEL	Domain Adaptive Ensemble Learning	MSS	Meaning Similarity Score
DANN	Domain-Adversarial Neural Network	NFV	Network Function Virtualization
DCI	Disentanglement, Completeness, Informativeness	NLP	Natural Language Processing
DG	Domain Generalization	NSC	Neural-Symbolic Communication
DNN	Deep Neural Network	OFDM	Orthogonal Frequency Division Multiplexing
DP	Differential Privacy	O-RAN	Open Radio Access Network
DRO	Distributionally Robust Optimization	OS	Operating System
DRL	Deep Reinforcement Learning	PSNR	Peak Signal-to-Noise Ratio
D-SAM	Domain-Specific Aggregation Module	QoE	Quality of Experience
E2E	End-to-End	QoS	Quality of Service
eMBB	Enhanced Mobile Broadband	RL	Reinforcement Learning
FID	Fréchet Inception Distance	RNN	Recurrent Neural Network
FLOPs	Floating Point Operations	RRM	Radio Resource Management
GAN	Generative Adversarial Network	ROUGE	Recall-Oriented Understudy for Gisting Evaluation
Grad-CAM	Gradient-weighted Class Activation Mapping	RSC	Representation Self-Challenging
GRL	Gradient Reversal Layer	SC	Semantic Communication
GPT	Generative Pretrained Transformer	SemCom	Semantic Communication
HARQ	Hybrid Automatic Repeat Request	SER	Semantic Error Rate
HMI	Human-Machine Interaction	SHAP	SHapley Additive exPlanations
IG	Integrated Gradients	SNR	Signal-to-Noise Ratio
IoT	Internet of Things	SSL	Self-Supervised Learning
JCAC	Joint Communication and Control	STM	Semantic Textual Similarity Measure
JSCC	Joint Source-Channel Coding	TCAV	Testing with Concept Activation Vectors
KB	Knowledge Base	URLLC	Ultra-Reliable Low-Latency Communication
KPI	Key Performance Indicator	V2X	Vehicle-to-Everything Communication
LIME	Local Interpretable Model-Agnostic Explanations	VLM	Vision-Language Model
LLM	Large Language Model	VR	Virtual Reality
LPIPS	Learned Perceptual Image Patch Similarity	XAI	Explainable Artificial Intelligence
LRP	Layer-wise Relevance Propagation	XR	Extended Reality

human-understandable explanations. By revealing which features, representations, or patterns drive particular outputs, XAI enables developers and end users to interpret, validate, and refine AI decisions more effectively. Within the context of semantic communication, explainability not only sheds light on how semantic representations are formed and processed but also clarifies why certain interpretations or reconstructions are selected under specific contextual conditions. The importance of such transparency is especially pronounced in domains such as healthcare diagnosis, autonomous transportation, and critical infrastructure supervision, where trustworthy and accountable AI behavior is indispensable [7–9].

Although the integration of XAI into semantic communication systems is strongly motivated by both practical demands and emerging research trends, existing studies have not yet examined this convergence in a truly comprehensive and systematic manner. Most prior investigations in semantic communication predominantly concentrate on performance-driven objectives, such as enhancing semantic accuracy, improving robustness against channel impairments, or optimizing transmission efficiency, while devoting comparatively limited attention to the interpretability of the underlying AI models. Consequently, the fundamental question of how to design semantic communication systems that are not only intelligent and adaptive but also transparent and understandable remains largely unresolved. This lack of explainability restricts the deployment of such systems in safety critical and socially sensitive applications, where accountability and user trust are essential. In response to this gap, this survey provides a structured and in-depth exploration of XAI techniques, architectural designs, and integra-

tion strategies specifically tailored to semantic communication. To the best of our knowledge, it constitutes the first dedicated effort to systematically examine how explainability can be embedded into semantic communication frameworks in order to enhance their transparency, reliability, and overall trustworthiness.

To further underscore the novelty and comprehensive scope of this survey, Table 1 presents the mapping the contributions of this survey against gaps identified in prior surveys on semantic communication and explainable AI (XAI). While prior surveys have explored individual aspects such as XAI techniques, application domains, or general 6G vision, they often fall short in addressing the full spectrum of explainability in semantic communication. Specifically, most works lack a structured taxonomy, do not propose dedicated semantic communication frameworks, and seldom address critical concerns related to trust, security, or real-time deployment. In comparison, our survey uniquely integrates all these dimensions offering a unified and in-depth perspective that bridges theoretical foundations with practical deployment challenges. This comparison not only highlights existing research gaps but also positions our contribution as the first to holistically address the intersection of XAI and semantic communication for future intelligent networks.

For the reader's convenience, the main abbreviations used in this paper are listed in Table 2, and the paper's structure is shown in Fig. 1.

By the end of this survey, readers will have gained a thorough understanding of how XAI is shaping the future of semantic communication, paving the way for systems that are not only more interpretable and adaptable but also capable of redefining the boundaries of human-computer interaction.

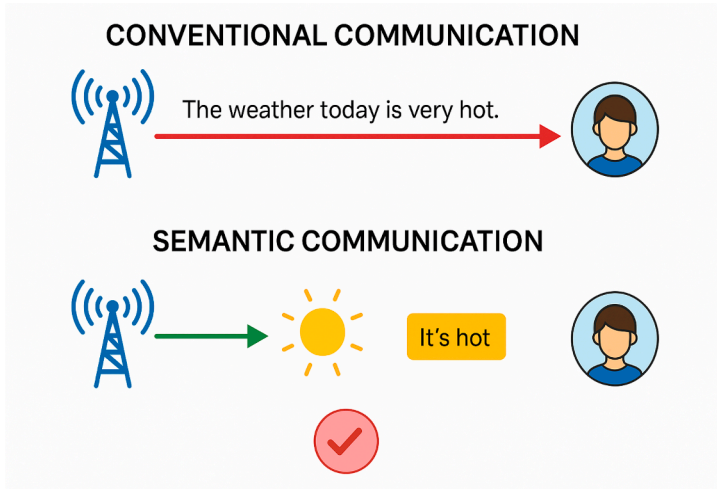


Fig. 2. Transmission via normal/conventional communication vs semantic communication • Transmitted message from BS or user “The weather today is very hot.” • This would simply involve transmitting the sentence as it is, word for word, using conventional encoding and decoding processes in a communication system. • This would focus on conveying the meaning or intent of the message rather than the exact words. For example, if the receiver already knows the context (e.g., it’s summer, and the region is known for high temperatures), the message might be compressed to something like “It’s hot” or even just a temperature value or code representing “very hot” weather. The receiver would then use this context to understand the full meaning of the message.

1.1. Theoretical foundations and technical rationale of semantic communication

Following the growing interest in enhancing interpretability within AI-driven communication systems, it is imperative to delve deeper into the theoretical and technical foundations of semantic communication. Traditional communication systems, largely built on Shannon’s model of information theory [45], have proven instrumental in achieving high-fidelity transmission through methods such as CDMA, OFDM, and MIMO [46–49]. These approaches are optimized to reduce noise, improve bit error rates, and ensure reliable delivery of signals over varying channel conditions. However, this framework is fundamentally agnostic to the meaning or relevance of the transmitted data. It assumes that all bits are equally important and treats every error as equally undesirable, regardless of whether it impacts the semantic interpretation of the message. In scenarios demanding real-time decision-making and context awareness, this model becomes insufficient.

Semantic communication introduces a transformative perspective by shifting the optimization criterion from symbol-level accuracy to semantic-level understanding. Rather than focusing on how accurately bits are transmitted, SemCom aims to ensure that the information understood by the receiver aligns with the sender’s intent. This concept is particularly critical in systems where the ultimate utility of a message depends not on exact replication but on correct interpretation [6,50]. In such cases, transmitting a semantically equivalent but lexically different message could be more efficient and equally effective as shown in Fig. 2.

The roots of this idea can be traced back to Warren Weaver’s three-tiered model of communication simply depicted in Fig. 3 [51], which segments communication into technical, semantic, and effectiveness levels. Weaver’s insights highlight that while the technical level focusing on encoding and transmission has been extensively studied and optimized, the semantic and effectiveness levels have largely been neglected. At the semantic level, the central question is whether the meaning intended by the sender is accurately reconstructed by the receiver. Weaver also noted that the greatest challenges and potential failures in communication lie in this semantic domain, where misunderstandings can arise

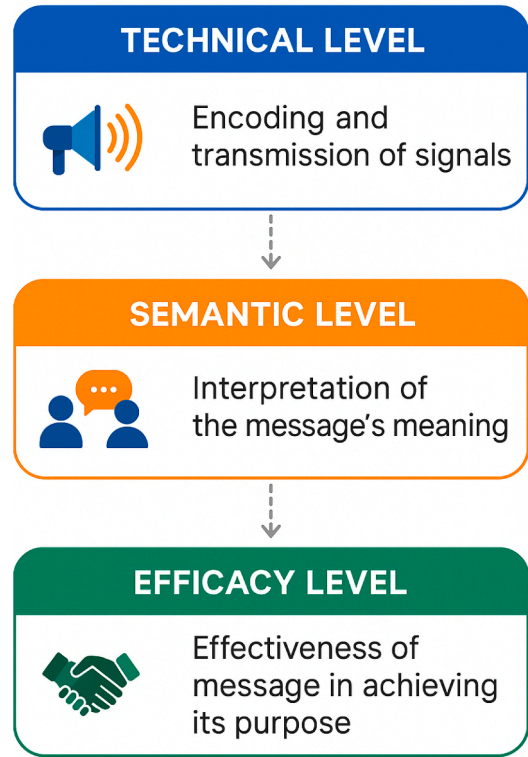


Fig. 3. The three major levels of semantic communications.

due to differences in interpretation, context, or background knowledge [52].

One of the reasons semantic communication did not gain traction earlier was the lack of computational tools capable of modeling and manipulating meaning. Shannon’s model dominated the evolution of communication technologies for decades because it offered a mathematically rigorous, tractable framework with practical applications [13]. However, the advent of AI, particularly deep learning and natural language processing, has provided the tools necessary to revisit and operationalize semantic-level communication. AI models can now perform context modeling, semantic parsing, and abstraction, thereby enabling machines to encode and decode messages based not just on their syntactic content but on their intended meaning [53,54].

At a technical level, semantic communication encompasses a suite of processes that fundamentally redefine how messages are constructed, transmitted, and interpreted as shown in Fig. 4. Key components of this framework include:

- **Context-Aware Encoding:** Leveraging the receiver’s prior knowledge and contextual information, the sender can minimize redundancy by transmitting only novel or uncertain content [55]. This approach leads to highly efficient communication by tailoring the message to the specific knowledge state of the recipient.
- **Semantic Compression:** Messages are compressed not in terms of bit count but in terms of semantic units. This allows for the transmission of only those elements that are critical for understanding [56]. For instance, semantic summarization techniques can distill the core intent of a long text into a concise representation.
- **Selective Filtering:** Irrelevant or low-utility information is filtered out during the preprocessing stage. This ensures that bandwidth is allocated to data segments that significantly influence the overall message interpretation [57].
- **Lossless Semantic Reduction:** Compression techniques are applied such that the receiver can reconstruct the intended meaning without

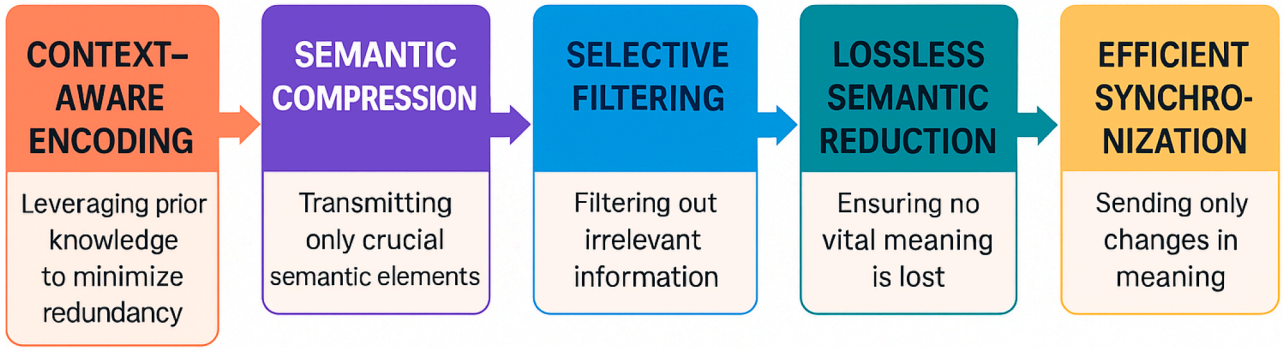


Fig. 4. Key components of semantic communication system.

Table 3
Impact of XAI on user interaction with different types of AI-enabled communication systems.

Ref	Aspect of XAI	Description	Example	Impact
[60]	Enhancing User Trust	Makes AI decisions transparent, increasing user trust.	A study on deep RL systems showing how XAI can make AI decisions clearer and more trustable.	Boosts user trust and acceptance.
[61]	Improving User Understanding	Helps users comprehend AI logic and decision-making processes.	XAI in diagnostic imaging helps radiologists understand AI's rationale in anomaly detection.	Enhances user comprehension and interaction with AI.
[62]	Personalization and Adaptability	Adapts responses to user preferences, enhancing satisfaction.	XAI in an ITS tailors explanations, improving relevance and engagement.	Increases user satisfaction and system engagement.
[63]	Error Correction and Feedback	Enables users to provide feedback on AI-generated content.	XAI in email classification allows feedback on misclassifications, improving accuracy.	Enhances system accuracy and user trust.
[64]	Evaluating System Confidence	Communicates system's confidence levels to users.	Study on industrial maintenance systems showing how transparency affects user confidence.	Improves user decision-making based on system reliability.
[65]	Addressing Ethical Concerns	Allows identification and correction of ethical issues.	Analysis of AI system flagging benign emails helps identify and correct biases.	Promotes responsible AI deployment and operational efficiency.

loss of critical semantic information, even if the lexical or syntactic structure changes [58].

- **Efficient Synchronization:** In dynamic environments, semantic communication allows for the transmission of only the changes in meaning or intent, rather than resending entire message structures. This drastically improves bandwidth utilization [59].

To quantify the efficiency gained through semantic encoding, let us define B_c as the bandwidth required in conventional communication and B_s as that required in semantic communication. Let α denote the fraction of a message that is semantically essential. The bandwidth saving S achieved through semantic communication can then be modeled as:

$$S = \frac{B_c - B_s}{B_c} = 1 - \alpha \quad (1)$$

A lower α indicates that much of the original message is either redundant or contextually recoverable, enabling significant bandwidth savings without compromising understanding.

Overall, semantic communication enables a redefinition of communication effectiveness. Rather than achieving reliability through redundancy and channel coding, it strives for efficiency by leveraging context, meaning, and intelligent inference. These foundations pave the way for integrating advanced interpretability mechanisms, such as XAI, which will be explored in subsequent sections. Together, SemCom and XAI offer a synergistic framework for building transparent, efficient, and intelligent communication systems suitable for the challenges of next-generation networks.

1.2. Enhancing semantic transparency: The role of XAI in intelligent and meaningful communication

The challenge of explainability within AI stems from the prevalent nature of AI tools functioning as “black-boxes,” making it difficult to construct analytical models that can verify their accuracy or provide

reasoning for its decisions in an understandable manner due to the opacity of black-box[66]. This lacking in explainability becomes particularly critical in scenarios where AI is implemented for real-time operations [67]. Unlike established practices in cellular networks and wireless standards, which have traditionally relied on methods that domain experts can analyze theoretically or simulate computationally to validate communication system components, the opacity of AI models poses a potential obstacle. Thus, achieving a comparable level of explainability in AI models is highly desirable. Addressing this challenge is paramount to foster trust, facilitate troubleshooting, and ensure the practical application of AI in the dynamic landscape of wireless communication networks [68].

AI-enabled communication techniques have shown promising capabilities and enhancements for resource allocation [69], beamforming [70], network management [71], security and privacy [72,73], edge computing [74], user experience[75], network planning and design [75,76] and semantic communications [21]. Even though many AI-enabled communication techniques have enhanced communication performance, this success is achieved by increasing the system complexity and employing a black-box. The black-box does not conduct operations transparently; instead, it generates output without giving clear explanations or reasoning for its decisions. Although these AI “black-box” models can achieve excellent levels of accuracy, their lack of interpretability may lead to complications, particularly in wireless communications which have a heterogeneous nature and also in real-time or practical applications where understanding the decision-making process is important [77]. To improve the comprehensibility of AI-enabled communication for humans and address the black-box problem by ensuring transparency in the decision-making process, we explore XAI, a specialized field within AI [78]. XAI has the potential to enhance network performance more efficiently. Furthermore, XAI can make substantial contributions to communication systems in several crucial ways as shown in Fig. 5. The Table 3 outlines the pivotal role of XAI in enhancing user interactions with AI. We categorize the benefits of XAI into



Fig. 5. Impact of XAI in communication systems.

six aspects that we further discuss with providing examples and impacts below.

1. Enhancing User Trust: In AI-enabled communication systems users may be doubtful or suspicious about relying on AI-generated outputs. XAI provides transparent explanations for AI decisions, increasing user trust by making the reasoning behind AI recommendations understandable.

The system model presented in [60] is a good example of making AI systems more transparent and enhancing user trust, particularly in the context of deep reinforcement learning (RL). The study's core revolves around an innovative XAI framework that not only sheds light on the AI's decision-making process through visual, narrative, and comparative analysis but also quantifiably boosts user trust and acceptance. By addressing the inherent 'black-box' nature of deep RL systems, the XAI framework provides a comprehensive suite of tools graphical representations, analogous environment performance assessments, and narrative explanations that collectively enhance user understanding of AI decisions. This study, which involved 18 participants, offers a comparative insight into user responses to AI systems with and without XAI support, revealing a significant uptick in user trust and acceptance with the XAI integrated approach. This user interface, equipped with explanatory features, aids users in grasping the AI's operational logic and its decision-making criteria, thereby nurturing user confidence and fostering a more intuitive interaction with AI technologies.

2. Improving User Understanding: XAI helps users comprehend the underlying logic and decision-making processes of AI systems. This understanding empowers users to interpret AI-generated content more effectively, leading to more meaningful and informed interactions.

As an example in [61], illustrates how XAI enhances the utility of AI in healthcare, particularly in diagnostic imaging. It describes a scenario where a radiologist uses an AI system to detect anomalies in X-ray images. In the absence of explainability, the AI merely highlights suspicious areas without elucidating the rationale, which may leave the radiologist in doubt about the AI's judgments. However, with XAI, the system not only identifies these areas but also provides justifications, such as detecting patterns consistent with lung nodules found in historical data. This added layer of explanation enables the radiologist to comprehend the AI's logic, assess the credibility of its findings, and make a more informed decision about integrating the AI's recommendations into the diagnostic process. Similarly, XAI will play an important role in several other fields as well for improving the user understanding.

3. Personalization and Adaptability: XAI enables communication systems to adapt their responses based on user preferences and require-

ments. By providing explanations, the system can tailor its communication to meet individual needs, enhancing user satisfaction.

The study [62], illustrates how XAI, through an explanation feature in an intelligent tutoring system (ITS), effectively adapts its communication to align with user preferences and requirements. The system customizes its interaction by offering detailed insights into why and how hints are provided, making it more relevant to individual learners. The positive impact on students' trust and perception of the hints, coupled with the modulation of the explanations' effectiveness by personal traits like conscientiousness and reading proficiency, underscores XAI's role in personalizing communication. This tailored approach meets specific user needs, thereby boosting user satisfaction and engagement with the system, embodying how XAI can enhance the adaptability and user-centricity of communication systems.

4. Error Correction and Feedback: XAI facilitates error detection and feedback in communication systems. Users can identify potential mistakes or biases in AI-generated content and provide corrective feedback, leading to improved accuracy and reliability.

For example, in a corporate environment, an AI-driven email classification system initially misclassifies essential emails as spam. To overcome this problem, the system in [63] incorporates an XAI framework, allowing users to provide feedback on misclassifications. The XAI system explains the AI's reasoning, enabling users to understand and trust its processes. This feedback adjusts and improves the AI model, enhancing its accuracy over time. As a result, the system more effectively categorizes emails, boosting user confidence and communication efficiency within the organization. This dynamic feedback loop exemplifies how XAI facilitates error correction and continuous learning, leading to a more reliable and user-centric AI system.

5. Evaluating System Confidence: With XAI, communication systems can convey their confidence levels in responses. Users can be informed when the system is uncertain about an answer, enabling them to make more informed decisions based on the AI's reliability.

The study [64], investigates XAI's impact on decision support systems within industrial maintenance and found that transparency levels such as black-box, grey box, and white box affect user confidence. The research aimed to show that XAI enhancements improve decision-making confidence. Users were given AI systems with varying transparency; black-box provided no explanations, grey box offered some insights, and white-box fully exposed its decision process. The outcomes highlighted that understanding AI reasoning through explanations, particularly in grey box and white box scenarios, boosts user trust, enabling more informed decisions in critical situations with significant consequences.

6. Addressing Ethical Concerns: Communication systems may encounter ethical challenges, such as biased or harmful content generation. XAI allows users to assess the system's behavior and identify potential ethical issues, leading to responsible and ethical AI deployment.

For example, when an AI system flags a benign email as phishing, XAI techniques can elucidate the rationale behind this decision by pinpointing the specific email characteristics that triggered the alert. Such insights can reveal if common but benign phrases or patterns inadvertently influenced the AI, leading to false alarms. This level of understanding empowers IT teams to fine-tune the AI system, reducing wrongful classifications and fostering trust in its judgements. Consequently, organizations benefit from a more reliable and discerning AI system that adeptly separates genuine threats from legitimate communications, thereby optimizing operational efficiency and safeguarding sensitive information[65].

Overall, XAI's contribution to communication systems can foster the QoS by promoting a better understanding of AI-generated content and enabling user feedback. XAI enhances the user experience and supports more effective and ethical communication between humans and AI systems. Though XAI in communication is still a new topic, it has vast potential to contribute to next-generation communication. Nevertheless, AI-enabled communication will play a vital role in

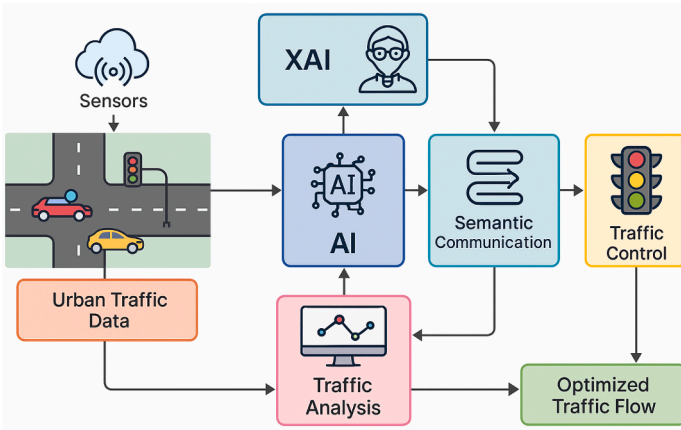


Fig. 6. A use case example of XAI-enabled SemCom: Smart urban traffic management.

developing and evolving future 6G communication systems. Notably, the ongoing development of 3GPP Release 20, which includes work items on AI/ML-driven designs and foundational studies for 6G, has attracted growing interest from both industry and academia seeking to contribute to the transformation of wireless communication [79].

From the lens of XAI within semantic communication, 3GPP Release 20 represents a forward looking milestone that aligns closely with the growing demand for transparency, adaptability, and intelligence in next-generation wireless systems. Building on the groundwork laid by Release 19, Release 20 is not a final specification but an evolving blueprint aimed at enhancing 5G Advanced and setting the stage for future 6G innovations. With a particular emphasis on AI/ML-native network architectures, integrated sensing, and non-terrestrial networks, Release 20 underscores the industry's commitment to embedding semantic awareness and interpretability at the core of communication design [9,80]. For semantic communication systems powered by AI, this evolution brings both opportunities and challenges. The focus on reducing latency, improving energy efficiency, and increasing network throughput is particularly critical for applications requiring real-time semantic inference such as immersive extended reality (XR), autonomous mobility, and time sensitive industrial operations [3,81]. However, as AI models take on greater roles in interpreting and managing semantic content, ensuring their decisions are explainable becomes essential for maintaining trust, safety, and regulatory compliance [6,8]. Moreover, features being explored in Release 20 such as network slicing, intelligent resource orchestration, and satellite terrestrial integration demand context-aware communication systems capable of dynamic, explainable decision making [7,32]. XAI will be indispensable in these scenarios, offering insights into how semantic models adapt across different service types, environments, and user profiles. Ultimately, Release 20 signals a shift not only toward greater intelligence in the network, but also toward accountable and interpretable AI-driven communication hallmarks of a trustworthy semantic infrastructure for the 6G era [82,83]. However, with these powerful capabilities of XAI we can build a robust trust among human and machine communication [84].

As we talk about XAI and semantic communication to build trust and enhance intelligence in the communication systems, it is important to note that the maturation of any technology is inevitably accompanied by a set of challenges that require careful consideration and targeted solutions. Within the realm of XAI, several constraints and hurdles demand attention, particularly in the context of communication systems. One noteworthy constraint revolves around the issue of interoperability within communication systems [85]. The intricate coexistence of diverse systems, devices, sensors, and equipment in communication systems presents a formidable challenge in the seamless exchange of loca-

tion and time dependent information [86]. This challenge manifests in the form of disparities in data formats, languages, data models, construction, and the intricate web of complex interrelationships among these components. The need for a standardized and interoperable framework is imperative to overcome these hurdles and establish a cohesive ecosystem within communication systems [87]. A second limitation pertains to the escalating complexity of XAI models. As the intricacy of these models increases, the inherent interpretability feature diminishes, presenting a considerable challenge in rendering the decision-making process transparent and comprehensible to users. Addressing this limitation requires innovative methodologies that balance model sophistication with interpretability, ensuring that users can grasp and trust the decisions made by AI systems even in the face of intricate models [88].

As an illustrative example, consider modern urban traffic management systems as shown in Fig. 6, where AI-enabled platforms routinely perform critical tasks such as forecasting traffic flow, detecting accidents, and dynamically coordinating traffic signals [89]. Despite these capabilities, practical deployments often suffer from inconsistencies in data representation and from the intricate interactions among heterogeneous AI models operating across different subsystems, which can introduce inefficiencies and limit overall system effectiveness [90,91]. A unified semantic communication framework, augmented with explainability mechanisms, offers a promising means to mitigate these challenges by enabling more coherent and interpretable information exchange among interconnected components, thereby improving transparency and fostering trust [92]. Importantly, the purpose of XAI in this context is not to replace the underlying predictive or control algorithms, but rather to expose the reasoning processes that lead to AI-driven outputs, such as congestion forecasts or adaptive signal control decisions, particularly when those outputs deviate from expectations. For example, if an AI system predicts an emerging traffic jam due to a suspected accident, XAI can clarify which features or patterns in the data contributed most strongly to this conclusion, allowing neighboring subsystems to understand and verify the basis of the prediction. This shared interpretability supports more coordinated actions, such as timely signal reconfiguration or the recommendation of alternative routes, grounded in understandable and consistent reasoning. A central challenge, however, lies in ensuring that such explanations are not only transparent but also sufficiently concise and actionable for use across distributed traffic control components. While AI concentrates on generating accurate predictions, XAI enables continuous system introspection by revealing the logic behind those predictions, which in turn supports refinement and performance improvement over time. Looking ahead to 6G networks, where AI will be entrusted with increasingly complex functions including resource orchestration and deep reinforcement learning based decision making, the role of XAI becomes even more critical for sustaining trust, accountability, and real-time operational efficiency [93].

Moreover, embedding XAI mechanisms into communication systems inevitably introduces additional layers of architectural and operational complexity. Effective solutions must therefore ensure that explainability is achieved without compromising system security, reliability, or computational efficiency, particularly on resource constrained edge devices. This requires carefully balancing advanced AI functionalities with the limited processing power, memory, and energy budgets available at the network edge. Consequently, the design of lightweight explanation models, along with robust security mechanisms that protect both data and explanations, becomes essential. Parallel efforts in algorithmic optimization and hardware aware implementation are also necessary to guarantee that XAI-enhanced models can execute efficiently within practical deployment environments [94].

Beyond issues of interoperability, model complexity, and deployment feasibility, XAI-enabled systems face fundamental challenges related to generalization. In this context, generalization refers to the ability of an AI model to maintain reliable performance and to produce meaningful explanations across previously unseen scenarios, domains, or operating conditions. This aspect is particularly important

in dynamic real-world environments, such as urban traffic networks, autonomous vehicular systems, smart manufacturing floors, large scale healthcare monitoring infrastructures, and intelligent energy distribution grids, where operating conditions evolve continuously and unpredictably. The difficulty arises because explanations learned under one distribution may not transfer seamlessly to another, leading to degraded interpretability even if predictive accuracy remains acceptable. For instance, an XAI model trained using data from a specific city or traffic pattern may struggle to generate coherent explanations when deployed in a different urban setting or under atypical conditions [95]. Ensuring that both predictions and their accompanying explanations generalize across contexts therefore remains an open research problem, and addressing it is crucial for establishing XAI-enabled semantic communication systems that are robust, dependable, and suitable for long-term real-world operation. Furthermore, it's important to underscore that the field of explainability itself is still in its early stages of development. Despite the progress made, there is a recognition that achieving comprehensive, reliable, and universally applicable explanations remains a complex challenge [96]. Acknowledging the infancy of explainability in AI is crucial for setting realistic expectations. While there have been significant strides in developing techniques for providing insights into AI decision-making processes, achieving a universally applicable and foolproof solution is a work in progress. Researchers are actively exploring diverse methodologies, from rule-based explanations to visualizations, to address the unique challenges associated with different AI applications [97].

In light of these identified limitations, further research is imperative to devise comprehensive solutions. Our in-depth study endeavor is pivotal, forming a synergy between AI-enabled semantic communication models and XAI techniques to address the dual challenges of model complexity and interpretability in AI systems. By integrating these methodologies, we aim to develop robust communication models that are transparent, interpretable, and adaptable across diverse environmental conditions. Our use case in healthcare diagnostics and urban traffic management as discussed above in this section exemplifies how this integration enhances trust and understanding in AI's decision-making, paving the way for broader applications. This endeavor is crucial for advancing AI systems that are not only technically sophisticated but also comprehensible and reliable, ensuring their transformative impact across various domains.

2. General architecture of XAI-enabled semantic communication

This section presents a modular and extensible architecture for AI-enabled semantic communication systems in which explainability is treated as a first-class design objective rather than a post-hoc add-on. The architecture is organized around seven tightly coupled functional modules, including the semantic encoder, semantic compression unit, channel encoder and physical layer, semantic decoder, XAI engine, user feedback loop, and explanation interface. Each module is designed not only to perform its core communication task, but also to expose interpretable signals that support transparency, traceability, and human understanding of semantic-level decisions.

By jointly considering semantic fidelity, communication efficiency, and explainability, the proposed architecture provides a structured foundation for building intelligent yet accountable communication systems. The role of each module and the associated interpretability mechanisms are discussed in detail in the following subsections. Fig. 7 illustrates the overall system architecture and highlights the interactions among the individual components, while Table 4 summarizes the corresponding input-output representations, computational and memory characteristics, and typical deployment locations (edge or cloud) for each module.

Beyond architectural design, this section also sets the stage for systematic evaluation. In particular, the performance metrics used to quantify semantic fidelity, explainability quality, and system-level overhead are discussed later in this section, followed by an end-to-end experimen-

tal roadmap that outlines representative tasks, datasets, baselines, and evaluation protocols for benchmarking XAI-enabled semantic communication systems under realistic wireless conditions.

- **Semantic Encoder:** The communication process initiates with an input message $M \in \mathcal{M}$, where $\mathcal{M}$ denotes a space comprising structured or unstructured data, including natural language, visual signals, or multimodal inputs. The primary objective at this stage is to extract the underlying meaning embedded in the input, rather than simply encoding syntactic or pixel-level information. This is achieved through a semantic encoder, a deep neural function defined as:

$$s = f_{\text{enc}}(M; \theta_{\text{enc}}) \quad (2)$$

Here, f_{enc} is a parameterized function implemented using architectures such as Transformers, BERT, or Vision Transformers, and θ_{enc} represents the learnable weights optimized via backpropagation. The output $s \in \mathbb{R}^d$ is a dense latent vector that captures the high-level semantic abstractions necessary for downstream compression, transmission, and reconstruction [98].

To ensure transparency in the semantic abstraction process, interpretability techniques are embedded within this stage. Feature attribution methods such as SHAP [99] and LIME [100] are employed to quantify the contribution of each input token or region to the generated representation s , offering valuable insights into the model's internal decision making. Furthermore, attention visualization methods including attention rollout, gradient based saliency, or Layer wise Relevance Propagation (LRP) can be used to reveal how contextual dependencies are distributed across the input [101,102]. These mechanisms not only enhance the interpretability of learned representations but also support model debugging and validation in real time scenarios.

- **Semantic Compression and Resource Optimization:** To enhance transmission efficiency in semantic communication systems, the high dimensional semantic vector $s \in \mathbb{R}^d$, produced by the encoder, is transformed into a compact representation $z \in \mathbb{R}^k$, where $k < d$, through a learnable compression function:

$$z = f_{\text{comp}}(s; \theta_{\text{comp}}) \quad (3)$$

Here, f_{comp} denotes the semantic compression model parameterized by θ_{comp} . This module is designed to retain essential semantic attributes while reducing redundancy and dimensionality, thereby lowering the communication overhead in terms of bandwidth, latency, and energy consumption. Compression strategies may include variational bottlenecks [103], entropy constrained neural encoders [104], quantization aware autoencoders, or task aware dimensionality reduction. These are typically optimized using semantic similarity metrics, rate distortion objectives, or downstream task performance [42].

Incorporating explainability into this stage enables a deeper understanding of how semantic relevance is preserved or lost during compression [105]. For instance, rule extraction methods such as FACET can be applied to interpret which latent dimensions of s are deemed critical and thus retained in z , and which are suppressed. Visualization techniques, such as semantic relevance heatmaps, may be used to highlight parts of the original input that significantly contribute to the compressed code. This provides interpretive transparency into the internal decision making of the model and supports diagnostics when semantic fidelity degrades [106].

Moreover, by exposing the inner workings of the compression process, explainable semantic compression allows developers and users to better understand the trade offs between information retention and resource constraints.

a. Latent-Manifold Interpretability of the Compressed Code z .

While attribution methods help explain how input features influence semantic representations, interpreting the compressed latent vector z requires concept-based and latent-space explainability

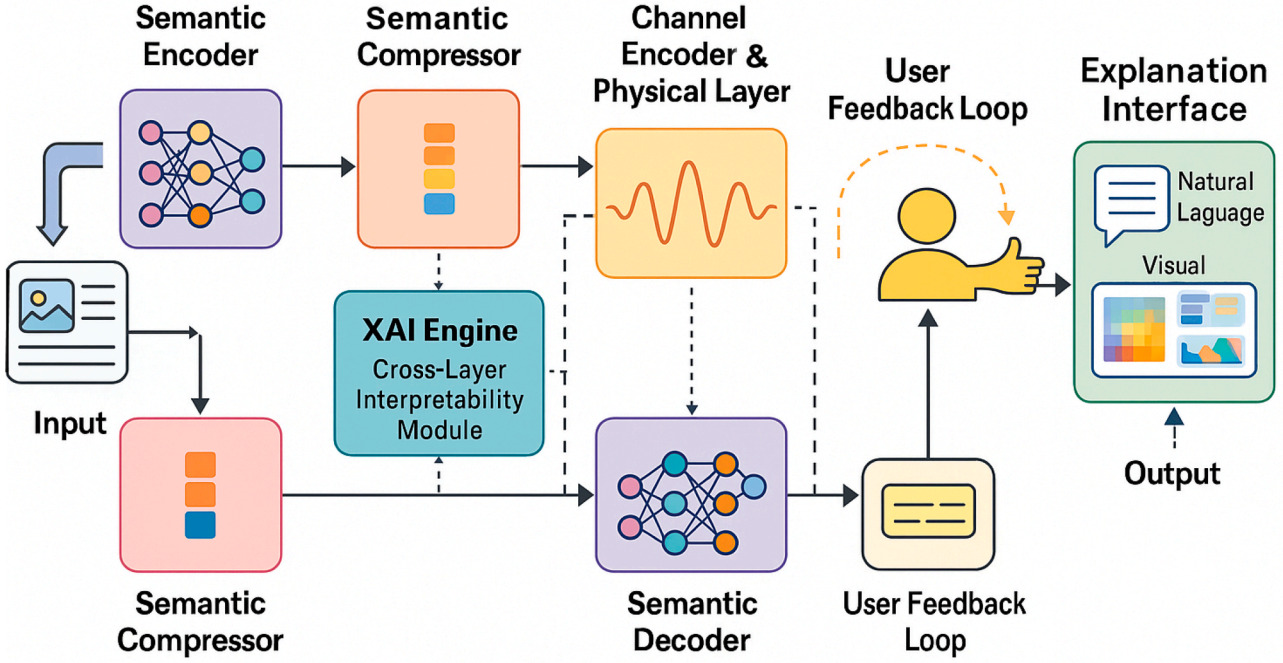


Fig. 7. General architecture of AI-enabled semantic communication highlighting key modules that facilitate interpretability through explainable encoding, adaptive transmission strategies, human-in-the-loop feedback, and cross-layer explanation synthesis.

Table 4

Illustrative module-level specification and resource budget for XAI-enabled semantic communication.

Module	Input → Output (Dimension)	Compute & Memory	Typical Placement
Semantic Encoder	$M = \{w_i\}_{i=1}^T \rightarrow s \in \mathbb{R}^d$	$O(T^2d)$; mem. $\propto P_{\text{enc}}$	Edge / Cloud
Semantic Compression	$s \in \mathbb{R}^d \rightarrow z \in \mathbb{R}^k$	$O(dk)$; mem. $\propto P_{\text{comp}}$	Edge
Channel Mapping	$z \in \mathbb{R}^k \rightarrow x \in \mathbb{C}^n$	$O(n)$; negligible mem.	Edge
Physical Channel	$x \rightarrow y = h \odot x + n$	–	Wireless Medium
Semantic Decoder	$y \rightarrow \hat{s} \in \mathbb{R}^d \rightarrow \hat{M}$	$O(T^2d)$; mem. $\propto P_{\text{dec}}$	Edge / Cloud
XAI Module	(model, activations) → E	$\eta_{\text{xai}} \times$ inference; $\eta_{\text{xai}} \approx 1-2$ (typ.), $\gg 10$ (worst)	Edge / Cloud
Adaptation / Feedback	$E, u \rightarrow \theta'$	Low–Moderate; mem. $\propto P_{\text{pol}}$	Cloud

Notation: $M = \{w_i\}_{i=1}^T$ is the source message of length T ; $s \in \mathbb{R}^d$ denotes the semantic embedding; $z \in \mathbb{R}^k$ ($k \ll d$) is the compressed latent representation; $x, y \in \mathbb{C}^n$ are transmitted and received channel symbols; h and n denote the channel coefficient and additive noise; $\hat{s}$ and $\hat{M}$ denote reconstructed semantic embedding and message, respectively; E denotes the generated explanation; u represents optional user feedback; θ' denotes updated model parameters. $P_{\text{enc}}, P_{\text{comp}}, P_{\text{dec}}, P_{\text{pol}}$ denote the number of learnable parameters per module, with parameter memory $\approx 4P$ bytes (FP32) or $2P$ bytes (FP16), while runtime memory also includes activation buffers that scale with intermediate tensor dimensions. η_{xai} captures the relative computational overhead of explanation generation compared to standard inference.

rather than purely input–output perturbation techniques. In particular, *Testing with Concept Activation Vectors (TCAV)* can be employed to associate human-interpretable semantic concepts (e.g., intent, object category, action, urgency) with directions in the latent space. A TCAV score then quantifies whether variations along a concept direction systematically increase or decrease a downstream semantic decision, enabling the interpretation of which semantic factors are encoded in specific regions of z .

Furthermore, *latent traversal and counterfactual decoding* can be applied by perturbing a single coordinate z_j (i.e., $z' = z + \delta e_j$) and observing the resulting change in the reconstructed message $\hat{M}$ or semantic utility. If small variations of z_j consistently induce changes in a specific semantic attribute, that coordinate can be associated with that attribute. Finally, *disentanglement-oriented metrics*, such as the Mutual Information Gap (MIG), are used to assess whether individual dimensions of z correspond to distinct semantic factors, or whether interpretability should instead be performed at the level of concept directions or latent subspaces rather than single coordinates.

- **Channel Encoder and Physical Layer Transmission:** Following semantic compression, the compact representation $z \in \mathbb{R}^k$ is processed

by the channel encoder to enable robust transmission over the physical medium. The channel encoder, denoted by f_{chan} , maps the compressed semantic vector into a signal space suitable for transmission:

$$x = f_{\text{chan}}(z; \theta_{\text{chan}}), \quad y = h(x) + n \quad (4)$$

Here, θ_{chan} represents the learnable parameters of the encoder, $x \in \mathbb{C}^n$ is the transmitted signal in complex baseband form, $h(\cdot)$ denotes the stochastic channel operator modeling fading or distortion (e.g., AWGN, Rayleigh, or Rician channels), and $n \sim \mathcal{N}(0, \sigma^2)$ represents additive white Gaussian noise (AWGN).

Unlike traditional communication systems, which employ explicitly defined modulation and coding schemes such as QAM, PSK, or OFDM, semantic communication systems may utilize neural encoders that implicitly learn optimal transmission strategies through end-to-end training. These learned schemes can adapt modulation depth, waveform structure, and coding redundancy based on the statistical properties of both the semantic input and the channel environment [107,108].

To ensure interpretability of these adaptive and often non-transparent strategies, explainability mechanisms are incorporated

into the channel encoding pipeline. For example, saliency analysis and relevance propagation can reveal which elements of the compressed semantic vector most influence the modulation behavior or symbol mapping. In systems employing reinforcement learning or policy gradient methods to dynamically adjust transmission parameters (e.g., power allocation, symbol shaping, or adaptive coding rates), the resulting policies can be analyzed using symbolic decision trees or heatmaps that expose policy logic under varying wireless channels [16].

Such explainability is crucial not only for verifying the correctness and efficiency of learned transmission strategies but also for diagnosing failure modes caused by unexpected channel behaviors or adversarial perturbations. Moreover, it enables system designers to evaluate robustness and generalizability across diverse communication scenarios.

- **Semantic Decoder:** At the receiver side, the principal objective is to accurately reconstruct the transmitted semantic content, despite the distortions introduced by noise, fading, and compression. This reconstruction process proceeds in two stages. First, the received signal $y \in \mathbb{C}^n$ is processed by a neural channel decoder f_{dechan} , which aims to reverse the physical layer transformations and recover the latent semantic representation $\hat{s} \in \mathbb{R}^d$:

$$\hat{s} = f_{\text{dechan}}(y; \theta_{\text{dechan}}) \quad (5)$$

Subsequently, the semantic decoder f_{dec} , parameterized by θ_{dec} , transforms the recovered semantic vector $\hat{s}$ into the reconstructed message $\hat{M} \in \mathcal{M}$:

$$\hat{M} = f_{\text{dec}}(\hat{s}; \theta_{\text{dec}}) \quad (6)$$

These two functions collectively serve the role of demodulation, error correction, and semantic reconstruction. The performance objective at this stage is not merely to minimize bit-level errors but to maximize semantic consistency between the original message M and the reconstructed message $\hat{M}$, typically measured using semantic similarity metrics, task-specific accuracy, or utility preservation [21].

To facilitate transparency in this reconstruction process, explainability techniques are integrated within both the channel and semantic decoders. Attention-based visualizations (e.g., using gradient weighted attention or Layer wise Relevance Propagation) help identify which portions of the received signal y or the intermediate vector $\hat{s}$ are most influential in determining the output $\hat{M}$. This allows practitioners to inspect whether the model prioritizes meaningful signal components or artifacts resulting from noise [101].

Moreover, counterfactual analysis can be employed to assess the sensitivity and reliability of the decoder. By introducing controlled perturbations to y or $\hat{s}$ and observing the corresponding variations in $\hat{M}$, one can infer the stability and semantic robustness of the decoding process [109].

- **XAI Engine - Cross Layer Interpretability Module:** The XAI engine constitutes a centralized interpretability framework that interfaces with all critical components of the semantic communication system. Rather than operating in isolation, the XAI engine is designed to continuously monitor, extract, and synthesize explanation signals across the semantic encoder, compressor, channel encoder, decoder, and user interaction modules. Its primary function is to ensure that both local (instance specific) and global (system wide) behaviors of the communication model are comprehensible, transparent, and aligned with human reasoning [110].

At the local level, the XAI engine employs model-agnostic explanation methods such as LIME [111], Integrated Gradients [112], and DeepLIFT [113] to elucidate the decision pathways underlying individual predictions or message transformations. These methods estimate the marginal contribution of input features such as specific tokens, image regions, or signal segments toward the system's output

at different layers. Such insights are essential for validating system outputs, especially in dynamic or critical environments [114].

For global interpretability, the engine may leverage statistical techniques such as SHAP (SHapley Additive exPlanations) [99] to quantify the average influence of input features across an entire dataset or operating domain. This enables practitioners to understand general patterns, biases, and priorities embedded in the learned model, such as over-reliance on particular features or underutilization of contextual semantics [115].

The engine also includes advanced visualization tools that support intuitive comprehension of internal representations. Techniques such as Grad-CAM [116], Layer-wise Relevance Propagation (LRP) [117], and attention flow visualizations [101] provide graphical mappings of where and how attention is distributed within the system, from input semantics to transmitted signals. These visual explanations are critical for debugging and also serve as pedagogical tools for human users interacting with the system [118].

Additionally, the XAI engine incorporates symbolic reasoning mechanisms to extract rule based representations from otherwise opaque neural decision boundaries. For example, decision trees or if-then logic can be derived from encoder outputs or compression decisions, offering concise, human readable summaries of model behavior.

• User Feedback and Adaptation Loop:

A defining characteristic of XAI-enabled semantic communication systems is the incorporation of a human-in-the-loop adaptation mechanism, through which end users can assess system outputs and influence subsequent model behavior. This feedback loop provides a practical bridge between model-centric optimization and user-centric objectives, enabling continual refinement of semantic representations in response to human preferences, contextual expectations, and task-specific requirements [119,120]. By involving users directly in the learning process, the system adapts not only to changing environments but also to evolving notions of semantic relevance and acceptability.

After a reconstructed message $\hat{M}$ is generated at the receiver, users may evaluate its semantic correctness, contextual relevance, or stylistic adequacy and provide explicit or implicit feedback. Such feedback is encoded into a user-defined loss function $\mathcal{L}_{\text{user}}$, and the system parameters are updated via gradient-based adaptation,

$$\theta_{t+1} = \theta_t - \eta_t \nabla_{\theta} \mathcal{L}_{\text{user}}(\theta_t), \quad (7)$$

where θ_t denotes the model parameters at update step t , η_t is the learning rate, and $\mathcal{L}_{\text{user}}$ captures discrepancies between system behavior and user expectations. Depending on the application, this loss may reflect semantic fidelity, intent alignment, stylistic conformity, or task-specific constraints.

a. Intermittent, delayed, and noisy feedback. In practical deployments, user feedback is rarely continuous. Instead, it arrives intermittently and may be delayed due to human response time, interaction latency, or limited user engagement. Let $b_t \in \{0, 1\}$ indicate whether feedback is available at update step t , with $\Pr(b_t = 1) = \rho$ denoting the feedback frequency, and let $\tau_t \in \{0, 1, \dots, \tau_{\max}\}$ represent the feedback delay. The resulting online adaptation can be expressed as

$$\theta_{t+1} = \theta_t - \eta_t b_t g_{t-\tau_t}, \quad g_{t-\tau_t} \triangleq \nabla_{\theta} \mathcal{L}_{\text{user}}(\theta_{t-\tau_t}; \xi_{t-\tau_t}), \quad (8)$$

where ξ_t captures stochasticity induced by user variability, contextual ambiguity, and channel effects.

A common abstraction for imperfect human feedback models the gradient as

$$g_t = \nabla_{\theta} \bar{\mathcal{L}}_{\text{user}}(\theta_t) + \beta_t + \epsilon_t, \quad (9)$$

where $\bar{\mathcal{L}}_{\text{user}}(\theta) = \mathbb{E}[\mathcal{L}_{\text{user}}(\theta; \xi)]$ is the expected user loss, β_t represents systematic bias (e.g., inconsistent preferences or annotation bias),

and ϵ_t is zero-mean noise with bounded variance. This formulation captures both random variability and structured inconsistencies commonly observed in human feedback signals.

b. Stability and convergence considerations. Under standard smoothness assumptions on $\bar{\mathcal{L}}_{\text{user}}$, delayed stochastic updates of this form can be interpreted as a closed-loop stochastic approximation process and are stable in expectation for sufficiently small step sizes. In particular, when $\bar{\mathcal{L}}_{\text{user}}$ is smooth and strongly convex, mean-square stability and convergence to a neighborhood of the optimum are ensured for bounded delay $\tau_t \leq \tau_{\max}$ if the step size satisfies

$$0 < \eta_t < \frac{c}{L(\tau_{\max} + 1)}, \quad (10)$$

where L is the Lipschitz constant of the gradient and $c \in (0, 1)$ is a conservative constant. In this regime, higher feedback frequency ρ accelerates adaptation but also increases sensitivity to noisy feedback, resulting in a steady-state error floor that scales with the feedback noise variance.

For nonconvex losses typical of deep semantic communication models, convergence to a stationary point in expectation can still be guaranteed under diminishing step-size schedules and bounded delay, although larger delays and sparser feedback slow the convergence rate. These observations emphasize the need to jointly balance feedback frequency, learning rate, and explanation reliability in practical deployments.

c. Stability evaluation under varying feedback conditions. Even without introducing additional end-to-end experiments, feedback-loop stability can be empirically characterized by monitoring the evolution of $\bar{\mathcal{L}}_{\text{user}}(\theta_t)$ and parameter variations $\|\theta_{t+1} - \theta_t\|$ over time for different feedback frequencies ρ , delay bounds $\tau_{\max}$, and injected feedback noise levels. Reporting confidence intervals across multiple random seeds provides a lightweight yet informative assessment of robustness under intermittent and noisy human corrections.

Explainability plays a critical role in improving both the stability and effectiveness of user-driven adaptation. By exposing the causes of semantic errors, transparent explanations enable users to provide more targeted and reliable feedback, thereby reducing variance and bias in the update signal. For example, attribution-based explanations can reveal overreliance on non-essential tokens or channel-induced artifacts, while counterfactual explanations illustrate how small semantic changes would alter system behavior [121]. Such insights support confidence-weighted updates, gradient smoothing, or clipping strategies that enhance robustness under noisy or inconsistent feedback.

User feedback can be incorporated within both supervised and reinforcement learning paradigms: annotated corrections directly refine semantic models, while user evaluations act as reward signals in policy optimization. In both cases, explainability improves convergence stability by aligning adaptation updates with human intent and task objectives [122,123].

- **Explanation Interface (Natural Language or Visual):** The final component of the XAI enabled semantic communication architecture is the explanation interface, which is responsible for transforming raw interpretability outputs often high dimensional, abstract, or model specific into formats that are accessible, intelligible, and actionable for human users. This interface serves as the primary point of interaction between the internal decision logic of the system and the diverse external stakeholders who must interpret, validate, or act upon its outputs [124].

One of the primary functions of the explanation interface is the generation of natural language explanations. Leveraging techniques from neural natural language generation (NLG), the system can produce concise textual justifications describing the rationale behind key decisions at various stages such as why specific features were

prioritized during semantic encoding, how compression choices affected transmission fidelity, or what semantic cues led to a particular reconstruction outcome. These explanations are grounded in attention weights, saliency maps, or attribution scores, and are articulated in user friendly language that supports non technical stakeholders [23].

Complementing these textual narratives are visual representations that offer more intuitive and interactive modalities for understanding system behavior. These include semantic heatmaps that highlight influential regions in input data, compression diagnostics that show retained versus discarded semantic features, channel behavior visualizations that indicate noise sensitivity or modulation characteristics, and decoder attention plots that illustrate how the system reconstructed meaning. Interactive dashboards may also allow users to explore explanation layers dynamically, compare outputs across different channel conditions, or interrogate decision boundaries using sliders and probes [124].

By presenting interpretability outputs in both textual and graphical formats, the explanation interface supports a wide spectrum of use cases from technical debugging and model auditing to end user trust calibration and regulatory compliance. It ensures that semantic communication systems are not only algorithmically sound but also human aligned, transparent, and accountable. This capability is essential in real world deployments where system decisions may have legal, ethical, or safety critical implications.

Latency and Overhead Considerations. The end-to-end latency incurred per semantic message can be approximated as

$$T_{\text{total}} = T_{\text{infer}} + T_{\text{xai}} + T_{\text{net}}, \quad (11)$$

where T_{infer} denotes the forward inference latency of the semantic communication pipeline, T_{xai} represents the additional overhead introduced by explanation generation, and T_{net} captures the communication delay between devices. For most gradient-based explanation methods, the explanation latency can be approximated as $T_{\text{xai}} \approx \eta_{\text{xai}} T_{\text{infer}}$, where η_{xai} depends on the selected XAI technique. In practice, lightweight attribution methods such as Grad-CAM typically incur $\eta_{\text{xai}} \approx 1$ –2, making them suitable for edge deployment, whereas more expensive methods such as integrated gradients or sampling based approaches may incur $\eta_{\text{xai}} \gg 10$ in worst case settings and are therefore better suited for execution on edge servers or cloud infrastructure. Typical edge deployments experience inference and explanation latencies on the order of a few to tens of milliseconds, while cloud-based execution benefits from higher computational throughput at the expense of additional network delay, which may range from several tens to hundreds of milliseconds depending on backhaul conditions [125]. This decomposition highlights the trade-offs between explainability richness, computational cost, and deployment location in XAI-enabled semantic communication systems.

In summary, the general architecture presented in this section integrates explainability as a foundational principle within each functional module of the semantic communication pipeline. By embedding interpretability mechanisms from encoding to decoding, from compression to channel transmission, and through to human interaction and feedback, the system transforms into a human-centric communication framework that prioritizes both semantic fidelity and transparency. The cross layer deployment of explainable AI techniques not only facilitates model introspection and performance debugging but also ensures alignment with human expectations and ethical AI principles. This holistic design is particularly advantageous for deployment in high stakes environments such as autonomous systems, medical diagnostics, and mission critical wireless communications where trust, reliability, and accountability are paramount. Ultimately, the synergy between semantic intelligence and interpretability sets the stage for the next generation of communication systems that are not only effective but also explainable, adaptive, and aligned with the goals of trustworthy AI.

2.1. Performance metrics for XAI-enabled semantic communication

Performance evaluation in AI-enabled semantic communication (SemCom) systems fundamentally differs from that of conventional communication systems, as the design objective shifts from bit-level reconstruction accuracy to the preservation of semantic meaning, task utility, and user trust. Consequently, performance metrics must jointly capture *semantic fidelity*, *explainability quality*, and the *resource overhead* introduced by explainable AI (XAI) mechanisms. While prior works commonly rely on classical NLP or task-oriented metrics, such measures alone are insufficient to assess the impact of explainability on semantic robustness and system efficiency. This section provides a unified and mathematically grounded formulation of performance and explainability metrics for XAI-enabled SemCom.

2.1.1. Semantic fidelity metrics

Let M denote a source message (text, image, or multimodal content), and let $\hat{M} = f(M; c, \theta)$ denote its reconstruction at the receiver under channel and system conditions c using model parameters θ . Semantic fidelity is defined as the expected preservation of meaning:

$$F_{\text{sem}} \triangleq \mathbb{E}_{(M, c) \sim \mathcal{D}} [U(M, \hat{M}; c)], \quad (12)$$

where $\mathcal{D}$ is the data channel distribution and $U(\cdot)$ is a bounded semantic utility function.

A generic instantiation of semantic utility combines semantic similarity and task success:

$$U(M, \hat{M}; c) = \alpha \text{sim}(\phi(M), \phi(\hat{M})) + (1 - \alpha) \text{Acc}_{\text{task}}(M, \hat{M}), \quad (13)$$

where $\phi(\cdot)$ is a learned semantic embedding function, $\text{sim}(\cdot, \cdot) \in [0, 1]$ is a normalized similarity measure (e.g., cosine similarity), and $\text{Acc}_{\text{task}} \in [0, 1]$ denotes task-level success (e.g., classification or decision accuracy). Both components are normalized to ensure a valid convex combination.

In text-based SemCom systems, BLEU and ROUGE scores can be interpreted as surface-form proxies for semantic similarity. However, as they primarily capture lexical overlap, they do not fully reflect meaning preservation under paraphrasing, semantic compression, or channel distortion. Therefore, embedding-based similarity and task-oriented accuracy are treated as primary indicators of semantic fidelity, while BLEU and ROUGE are retained as secondary baselines for comparability with prior studies.

2.1.2. Explainability metrics for semantic communication

Let $E(M)$ denote the explanation generated for input M (e.g., attribution map, concept relevance vector, or rule-based rationale). We define a unified explainability score as:

$$\mathcal{X}(M) \triangleq w_f \text{Faith}(M) + w_s \text{Stab}(M) + w_c \text{Comp}(M) + w_{\text{cal}} \text{Cal}(M), \quad (14)$$

where $w_i \geq 0$, $\sum_i w_i = 1$, and all components are normalized to $[0, 1]$.

2.1.3. (A) faithfulness:

Faithfulness measures whether the explanation captures features that causally influence semantic outcomes. Let $S_k(M)$ denote the set of top- k features (or latent dimensions) identified by $E(M)$. Faithfulness can be quantified via a deletion based semantic utility drop:

$$\text{Faith}(M) = \frac{1}{K} \sum_{k=1}^K \frac{U(M, \hat{M}; c) - U(M \setminus S_k, (\widehat{M \setminus S_k}); c)}{U(M, \hat{M}; c)}, \quad (15)$$

where $U(\cdot)$ is the semantic utility defined in (13).

2.1.4. (B) stability:

Stability captures robustness of explanations to semantics preserving perturbations such as paraphrasing, benign data augmentation, or mild channel variations:

$$\text{Stab}(M) = 1 - \mathbb{E}_{\tilde{M} \sim \mathcal{P}(M)} \left[\frac{d(E(M), E(\tilde{M}))}{d_{\text{max}}} \right], \quad (16)$$

where $\mathcal{P}(M)$ denotes the set of semantic preserving perturbations, $d(\cdot, \cdot)$ is a distance between explanations, and d_{max} ensures normalization.

2.1.5. (C) compactness:

Compactness quantifies explanation sparsity, which is critical under resource constraints. For attribution based explanations $e(M) \in \mathbb{R}^d$, compactness is defined as:

$$\text{Comp}(M) = 1 - \frac{\|\mathbb{1}(|e(M)| \geq \tau)\|_0}{d}, \quad (17)$$

where τ is a relevance threshold and d is the feature or latent dimension.

2.1.6. (D) calibration:

Calibration measures alignment between explanation confidence and actual semantic reliability. Using an expected calibration error (ECE) formulation:

$$\text{Cal}(M) = 1 - \sum_{b=1}^B \frac{|I_b|}{N} |\text{acc}(I_b) - \text{conf}(I_b)|, \quad (18)$$

where I_b denotes samples in confidence bin b , $\text{acc}(\cdot)$ is empirical semantic success, and $\text{conf}(\cdot)$ is explanation confidence.

2.1.7. Concept-based and latent-space explainability

To interpret compressed semantic representations, concept based metrics such as Testing with Concept Activation Vectors (TCAV) quantify sensitivity to human defined concepts:

$$\text{TCAV}_C(M) = \frac{1}{|\mathcal{T}|} \sum_{i \in \mathcal{T}} \mathbb{1}(\nabla_a f_i(a) \cdot v_C > 0), \quad (19)$$

where a is an internal activation, v_C is the concept vector, and f_i is the task specific logit.

Latent space interpretability can further be evaluated using disentanglement metrics such as the Mutual Information Gap (MIG):

$$\text{MIG} = \frac{1}{K} \sum_{k=1}^K \frac{I(z_{j(1)(k)}; v_k) - I(z_{j(2)(k)}; v_k)}{H(v_k)}, \quad (20)$$

where z_j denotes latent dimensions and v_k represents semantic factors when available.

2.1.8. Classical baseline metrics

For comparability with existing SemCom literature, classical baseline metrics such as BLEU, ROUGE, task accuracy, F1-score, and perplexity are provided in Table 5. These metrics provide reference points for surface-level similarity, linguistic fluency, and task success but do not assess explanation quality, causal faithfulness, or stability. Accordingly, they are treated strictly as baselines rather than comprehensive indicators of semantic fidelity or explainability.

2.1.9. Trade-off between semantic fidelity and explainability

Incorporating explainability into semantic communication systems inevitably introduces additional computational, latency, and bandwidth overhead. As a result, system design must balance the preservation of semantic fidelity against the cost of generating and conveying explanations. This balance can be formalized through a joint optimization framework that explicitly captures the competing objectives:

$$\min_{\theta} \lambda_{\text{sem}} \mathcal{L}_{\text{sem}} - \lambda_{\mathcal{X}} \mathbb{E}[\mathcal{X}(M)] + \lambda_o C_{\text{xai}}, \quad (21)$$

where $\mathcal{L}_{\text{sem}}$ denotes the semantic loss measuring meaning preservation, $\mathcal{X}(M)$ represents an explainability score associated with message M , and C_{xai} captures the resource overhead induced by explanation generation (e.g., latency, bandwidth, or energy). The weighting coefficients λ_{sem} , $\lambda_{\mathcal{X}}$, and λ_o control the relative importance of semantic performance, interpretability, and system cost. This formulation naturally supports multi-objective analysis, enabling designers to explore trade-offs rather than optimizing a single metric in isolation.

Pareto frontier and ablation-based analysis: To make the trade-off explicit, we define the semantic utility as $F_{\text{sem}}(\theta) \triangleq -\mathcal{L}_{\text{sem}}(\theta)$ and the explainability burden as $C_{\text{xai}}(\theta)$, which aggregates the additional

Table 5
Summary of baseline performance parameters in AI-enabled semantic communication.

Metric	Mathematical Expression	Description
Precision	$P = \frac{TP}{TP+FP}$	Fraction of retrieved samples that are relevant. High precision indicates fewer incorrect semantic outputs.
Recall	$R = \frac{TP}{TP+FN}$	Fraction of relevant samples that are successfully retrieved. High recall indicates better semantic coverage.
F1-Score	$F_1 = 2 \cdot \frac{P \cdot R}{P+R}$	Harmonic mean of precision and recall, balancing false positives and false negatives.
Accuracy	$\mathcal{A} = \frac{TP+TN}{TP+TN+FP+FN}$	Ratio of correctly predicted samples to all samples; commonly used in semantic classification tasks.
Mean Average Precision (mAP)	$mAP = \frac{1}{Q} \sum_{q=1}^Q \int_0^1 P_q(r) dr$	Average precision over recall levels and queries; widely used in semantic retrieval and ranking.
BLEU Score	$BLEU = BP \cdot \exp\left(\sum_{n=1}^N w_n \log p_n\right)$	Measures n-gram overlap between generated and reference text. BP penalizes short outputs; p_n is n-gram precision.
ROUGE Score	$ROUGE_n = \frac{\sum \text{Count}_{\text{min}(n, \text{gram})}}{\sum \text{Count}(n\text{-gram})}$	Measures n-gram recall between generated and reference summaries; commonly used in summarization tasks.
Perplexity	$PPL = \exp\left(-\frac{1}{N} \sum_{i=1}^N \log P(y_i y_{<i})\right)$	Measures uncertainty of a language model. Lower values indicate more fluent and confident semantic generation.
Embedding Similarity	$\cos(\theta) = \frac{\mathbf{w}_1 \cdot \mathbf{w}_2}{\ \mathbf{w}_1\ \ \mathbf{w}_2\ }$	Cosine similarity between semantic embeddings, indicating closeness in latent semantic space.
Semantic Similarity	$\text{sim}(\mathbf{u}, \mathbf{v}) = \frac{\mathbf{u} \cdot \mathbf{v}}{\ \mathbf{u}\ \ \mathbf{v}\ }$	Measures alignment between encoded and reconstructed semantic representations.

Notation: TP / TN / FP / FN denote true positives, true negatives, false positives, and false negatives, respectively. Q is the number of queries. $P_q(r)$ is precision at recall r for query q . y_i is the i -th token and $y_{<i}$ its preceding context. Bold symbols denote vector representations.

Table 6

Illustrative Pareto-style ablation points showing the impact of explanation strength α_{xai} on semantic utility and overhead (replace with measured values in the final revision).

α_{xai} Setting / XAI Method	F_{sem}	ΔT_{xai} (ms)	ΔB_{xai} (kb/msg)	Observed Effect
None ($f_{\text{xai}}=0$), –	0.88	0.0	0.0	Lowest cost
Low (top- $K=5$, $f_{\text{xai}}=0.1$), IG / Grad-CAM	0.88	1.6	0.4	Stable explanations
Medium (top- $K=10$, $f_{\text{xai}}=0.25$), IG / Grad-CAM	0.87	4.2	1.1	Better debugging
High (top- $K=20$, $f_{\text{xai}}=0.5$), IG / LRP	0.86	8.9	2.3	Richer explanations
Very High ($f_{\text{xai}}=1.0$), IG / LRP	0.85	16.7	4.6	Edge limits bind
Sampling-based (top- $K=10$, $f_{\text{xai}}=0.25$), LIME / SHAP	0.87	38.0	1.1	Offline preferred

Notation: α_{xai} controls explanation strength (e.g., frequency f_{xai} , granularity/top- K , or resolution). F_{sem} denotes semantic utility. ΔT_{xai} and ΔB_{xai} denote added latency and bandwidth per message due to explanation generation or transmission. Illustrative mean values; final version should report 95% confidence intervals and paired significance tests.

latency, bandwidth, or energy required for explanations. A parameter setting θ_a is Pareto-optimal if there exists no alternative θ_b such that $F_{\text{sem}}(\theta_b) \geq F_{\text{sem}}(\theta_a)$ and $C_{\text{xai}}(\theta_b) \leq C_{\text{xai}}(\theta_a)$, with at least one inequality being strict. Pareto-optimal points therefore characterize operating regimes where improvements in explainability or semantic fidelity cannot be achieved without incurring additional system cost.

In practice, Pareto frontiers can be obtained through lightweight ablation studies by sweeping an *explanation strength* parameter α_{xai} . This control variable may correspond to explanation frequency, attribution granularity (e.g., top- K features), or spatial/temporal resolution of explanations. For each setting of α_{xai} , the resulting pair $(F_{\text{sem}}, C_{\text{xai}})$ is reported to visualize the trade-off surface.

Let $f_{\text{xai}} \in [0, 1]$ denote the fraction of transmitted messages for which explanations are generated, and let $S_E(\alpha_{\text{xai}})$ represent the payload size of an explanation under granularity α_{xai} . The additional bandwidth consumed by explainability can then be approximated as

$$\Delta B_{\text{xai}} \approx f_{\text{xai}} \frac{S_E(\alpha_{\text{xai}})}{T_{\text{msg}}}, \quad (22)$$

where T_{msg} is the message transmission interval. Similarly, if explanation generation introduces a relative computational factor $\eta_{\text{xai}}(\alpha_{\text{xai}})$ compared to baseline inference, the added per-message latency can be expressed as

$$\Delta T_{\text{xai}} \approx f_{\text{xai}} (\eta_{\text{xai}}(\alpha_{\text{xai}}) - 1) T_{\text{infer}}, \quad (23)$$

with T_{infer} denoting the inference latency of the underlying semantic communication model.

These expressions provide a concrete and reproducible methodology for quantifying how richer or more frequent explanations impact

bandwidth and latency budgets, and how such overheads propagate to downstream task performance under fixed resource constraints. As such, Pareto-based analysis offers a principled mechanism for selecting operating points that balance semantic fidelity, explainability quality, and system efficiency in XAI-enabled semantic communication systems. To make the above trade-off concrete without introducing new experimental claims, Table 6 presents representative operating points that illustrate how increasing explanation strength is expected to impact semantic utility and system overhead in practice.

Taken together, the above analysis suggests that evaluation of XAI-enabled semantic communication systems should be grounded in semantic fidelity as the primary performance objective, assessed through embedding-based similarity measures and task-level utility rather than surface-level lexical overlap alone. Classical metrics such as BLEU and ROUGE remain useful for continuity with prior work, but should be interpreted strictly as secondary reference baselines rather than comprehensive indicators of meaning preservation or interpretability. Explainability must be evaluated explicitly using criteria such as faithfulness, stability, compactness, and calibration in order to capture the reliability and practical value of generated explanations. Equally important, the resource overhead introduced by explainability mechanisms should be reported transparently in terms of latency, energy consumption, and bandwidth usage, particularly for edge-constrained and real-time deployments. Finally, both semantic performance and explanation quality should be stress-tested under semantics-preserving perturbations, including paraphrasing and channel variation, to ensure robust and dependable operation in dynamic wireless environments. However, when comparing XAI settings, we recommend reporting Pareto curves

Table 7
End-to-end experimental roadmap for evaluating XAI-enabled semantic communication systems.

Testbed	Task and Modality	Representative Datasets	Channel Conditions	Baseline Systems	Evaluation Metrics and Statistical Reporting
TB-1	Text semantic communication (reconstruction / intent classification)	AG News, SST-2, or SNLI (with domain-shifted split)	AWGN and Rayleigh fading; SNR sweep; paraphrasing noise	Conventional source-channel coding; SemCom without XAI; SemCom with lightweight XAI (IG/Grad-CAM); SemCom with sampling-based XAI (LIME/SHAP)	Semantic fidelity (embedding similarity, task accuracy/F1); BLEU/ROUGE as secondary baselines; explainability faithfulness and stability; latency, energy, and memory; Pareto curves over explanation strength (α_{xai}) vs. overhead ($\Delta\text{latency}/\Delta\text{bandwidth}$) with 95% confidence intervals
TB-2	Vision semantic communication (classification or captioning)	CIFAR-10, ImageNet subset, MS-COCO or Flickr30k	AWGN and fading; packet loss; compression stress	Same as TB-1, emphasizing visualization-based XAI (LRP, Grad-CAM)	Task accuracy or caption quality; semantic similarity; explanation localization consistency; inference and explanation overhead; Pareto analysis over α_{xai} with confidence intervals across runs
TB-3	V2X semantic communication (intent or maneuver classification)	INTERACTION, NGSIM, or nuScenes intent labels	Time-varying fading; Doppler; mobility-induced distribution shift	Feature-based baselines; SemCom without XAI; SemCom with attribution-based XAI	Intent accuracy/F1; robustness under mobility and channel variation; explanation stability; edge vs. server latency profiling; Pareto trade-offs between explainability overhead and task performance with statistical testing

by sweeping α_{xai} (or $(\lambda_{\text{sem}}, \lambda_{\text{xai}})$) and plotting semantic utility against explainability overhead with confidence intervals across random seeds.

2.2. End-to-end experimental roadmap for XAI-enabled semantic communication

Although this survey primarily focuses on architectural design, explainability techniques, and evaluation metrics for semantic communication, their practical relevance ultimately hinges on end-to-end experimental validation. To bridge this gap, we outline a concise yet technically grounded experimental roadmap that translates the proposed XAI-enabled semantic communication framework into reproducible evaluation pipelines. Rather than introducing new empirical results, the purpose of this roadmap is to guide future implementations by identifying representative tasks, datasets, baselines, and evaluation protocols that enable systematic and fair benchmarking. Table 7 summarizes the proposed testbeds and illustrates how semantic fidelity, explainability quality, and system-level overhead can be jointly evaluated under realistic wireless conditions.

From a system perspective, experiments should explicitly model wireless impairments such as additive noise, fading, and mobility, enabling evaluation under varying signal-to-noise ratios and channel dynamics. For text- and image-based semantic communication tasks, this may involve transmitting learned semantic embeddings over Rayleigh or Rician fading channels, while V2X or XR-oriented scenarios may further incorporate latency constraints and bursty traffic patterns. In all cases, explainability mechanisms should be evaluated not only in isolation but also in conjunction with semantic reconstruction performance, allowing the interaction between semantic compression and explanation stability to be observed.

Across all test scenarios, experiments should be conducted over multiple random seeds to account for both training stochasticity and channel variability, with results reported using confidence intervals to quantify statistical uncertainty. When comparing explainable and non-explainable semantic communication baselines, paired statistical tests are recommended to isolate the impact of explainability mechanisms from architectural or optimization effects. In addition to semantic fidelity and task success, system-level metrics such as end-to-end latency, energy consumption, and bandwidth usage should be explicitly reported, as explainability modules introduce additional computation and memory access that may affect real-time operation.

To assess deployment feasibility, micro-benchmarking of inference and explanation generation latency should be performed on representative edge platforms or hardware accelerators. In practice, lightweight gradient-based explanations are expected to incur modest overhead compatible with on-device execution, whereas sampling-based or op-

timization driven explanation methods may impose significantly higher computational cost and are therefore better suited to cloud-assisted or offline analysis. Explanation quality itself can be validated through perturbation based faithfulness tests, stability under semantics preserving transformations, and where applicable controlled human evaluation to assess interpretability and usefulness. For each testbed in Table 7, Pareto frontiers can be obtained by sweeping explanation strength (e.g., Grad-CAM vs. IG vs. SHAP, or varying top- K /resolution and explanation frequency) and jointly reporting semantic fidelity, explanation quality, and overhead (latency/energy/bandwidth) under fixed resource budgets.

This evaluation methodology aligns with experimental practices commonly adopted in reinforcement learning based network optimization, V2X communication, and intelligent resource management studies, where multi-seed analysis, statistical rigor, and system-level constraints are explicitly considered [126]. By providing a technically grounded and reproducible experimental blueprint, this roadmap aims to facilitate objective assessment and accelerate the transition of explainable semantic communication systems from conceptual designs to practical wireless deployments.

3. State of the art AI and XAI enabled semantic communication

As semantic communication becomes increasingly vital for the future of intelligent wireless systems, a growing body of research has sought to explore its design principles, enabling technologies, and practical applications. These studies have primarily focused on the efficiency of message compression, the design of semantic encoders and decoders, and system level optimizations to reduce bandwidth and latency [6,127,128]. Semantic communication is often presented as a transformative advancement over traditional communication systems, particularly in the context of 6G, offering significant improvements in relevance aware data transmission, human-machine interaction, and goal-oriented services [80,129].

However, the literature reveals a consistent emphasis on performance gains while often sidelining the question of interpretability. The semantic models used in these systems typically based on deep neural networks, transformers, or generative models act as opaque black-boxes that are difficult to validate, debug, or trust, especially in mission critical or real-time applications [8,77]. This lack of transparency introduces significant limitations in deploying semantic communication across sensitive sectors such as healthcare, autonomous systems, and industrial automation, where understanding AI-driven decisions is essential for safety and accountability [9,92].

While several surveys have been published on XAI in wireless networks and IoT [82,83,130], and others have examined semantic communication in terms of architectures and applications [19,53,127], none

offer a dedicated or comprehensive treatment of explainability within the semantic communication paradigm. Existing reviews tend to address explainability in a general sense or isolate it to non-semantic communication use cases, leaving a critical knowledge gap unaddressed. Moreover, a closer inspection of these studies reveals that explainability is often framed as a post-hoc consideration introduced after the model has already been trained and deployed rather than as an integral part of the semantic system design. This reactive approach fails to consider how XAI methods can be integrated at the architectural level to enhance interpretability throughout the semantic communication pipeline.

In response to these limitations, our work aims to fill this gap by presenting a focused analysis of explainable AI in semantic communication. We examine how XAI techniques have been applied or can be applied to improve transparency, facilitate trust, support system diagnostics, and promote user centric design in semantic communication. The remainder of this section categorizes and analyzes the literature in terms of foundational studies, recent advancements, and technical challenges, offering a systematic perspective on the current state and future directions of XAI in semantic communication.

3.1. From black-box to glass-box: Explainability trends in communication systems

Explainability in communication systems refers to the ability to make complex concepts, ideas, or information understandable and transparent to the intended receiver [131]. In the context of AI and machine learning, explainability is essential for enabling effective human-machine interactions, especially when AI-enabled communication systems are involved in decision making processes. When AI-enabled communication systems provide outputs or recommendations, explainability ensures that the reasoning behind those outcomes can be conveyed to users in a clear and comprehensible manner [132]. This transparency is crucial for building trust in AI-enabled communication systems and helping users understand why a certain decision was made or a particular action was suggested [82,133]. In various fields, such as natural language processing [134], computer vision [135,136], and predictive analytics [137], explainability becomes vital to ensure that AI-enabled insights can be communicated to stakeholders, experts, or non-experts, without relying solely on technical jargon or complex algorithms. Overall, the goal of explainability in communication systems is to bridge the gap of keeping transparency in decision making between the technical processings of AI-enabled communication systems and the human understanding, fostering trust, accountability, and user acceptance of AI-enabled technologies. In Fig. 8, we present a simple comparison of XAI and Non XAI communication system. The XAI-enabled communication system consists of three main components: the machine learning processings (represented as a black-box model), the explainability model, and the explanation generation module.

1. Machine Learning Processings (Black-Box Model): The machine learning processings represents the black-box model, which is typically a complex, opaque model. It takes input data and performs the AI tasks, such as classification, prediction or recommendation. However, the inner workings/processings of the model are not easily interpretable, making it challenging to understand why it arrives at specific predictions or decisions [78]. The lack of interpretability in these models can pose significant challenges, especially when decisions need to be justified or understood by humans

2. Explainable Model: The explainable model is an intermediary between the machine learning process and the explanation generation module. Its purpose is to extract interpretable insights from the predictions made through machine learning processes. The explainable model analyzes the internal structure of the black-box model and identifies the critical features or patterns that influence its output [138].

3. Explanation Generation Module: This block is responsible for creating human understandable explanations based on the insights pro-

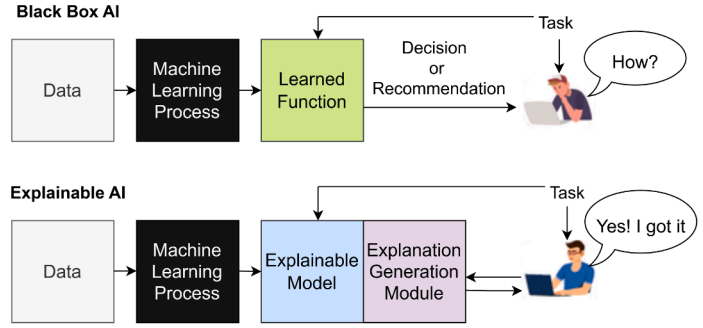


Fig. 8. Simple representation of Black-box AI and explainable AI model.

vided by the explainable model. It takes the output from the explainable model and translates it into human readable explanations, such as feature importance scores, decision rules, or visualization of decision making processes [78]. While the black-box AI model in Fig. 8 processes input data based on its internal architecture, it provides no insight into how decisions are made. This lack of interpretability stands in contrast to the XAI-augmented system, which offers transparent reasoning through explainability and explanation generation modules [15,139].

3.2. Literature insights on the advancements of semantic communication

Following the discussion on explainability in AI-enabled systems, it becomes imperative to assess how recent advancements in semantic communication (SemCom) have responded to the demand for more transparent, adaptive, and intelligent communication mechanisms. SemCom, by design, shifts focus from transmitting raw bits to conveying meaningful content, but the challenge of interpretability persists particularly as AI models used in encoding and decoding become increasingly opaque. While many studies have explored the architectural and algorithmic advancements in semantic communication such as joint source channel coding, attention based encoders, and transformer based semantic encoders their treatment of explainability is often superficial or absent. A few recent efforts have begun to incorporate explainability features either by using post-hoc interpretation methods or by embedding transparency into model design [21,138]. These early integrations suggest promising directions for enhancing human-machine communication reliability and trustworthiness. To build a clearer picture of current progress, we analyze key SemCom contributions across various dimensions: types of input features, AI methodologies adopted, performance metrics evaluated, and whether any level of interpretability is addressed. Table 8 offers a structured summary of these recent works, which we elaborate upon in the following discussion to identify trends, gaps, and future opportunities at the intersection of semantic communication and explainable AI.

In [140], the authors introduced the Task-Oriented Semantic Communications with Semantic Reconstruction (TOSC-SR) scheme, a system designed to preserve the semantic content of images while optimizing performance for various AI tasks. TOSC-SR addresses the problem through an extended rate-distortion optimization, which includes semantic information relevant to AI tasks. This system employs joint source and channel coding (JSCC) to support multiple AI tasks simultaneously. By using rate-distortion optimization (RDO), the system aims to find the best balance between transmission rate and the amount of distortion or error introduced. Experimental results demonstrate that TOSC-SR outperforms traditional communication methods like JPEG and JPEG2000, as well as other deep learning-based multi-task communication systems. This innovative model effectively combines rate-distortion optimization, semantic information preservation, and support for AI tasks to improve image semantic communications.

Table 8

Proposed taxonomy of opportunities: Mapping representative SemCom systems to suitable XAI classes and explanation targets.

Study	Input Modality	SemCom Layer / Sub-problem	Model Type	Explanation Target	Applicable XAI Paradigm
[140]	Image	Semantic Source Encoding	CNN	Salient regions in semantic features	Attribution (Grad-CAM, IG)
[141]	Image	Semantic Compression	GAN	Latent code importance	Disentanglement, Latent Traversal
[130]	Voice/Speech	Semantic Compression	Stochastic Encoder	Semantic bits retained/dropped	Feature Importance, Surrogate Models
[142]	Images	Semantic Reconstruction	ResNet	Pixel/region contribution to task	Grad-CAM, LRP
[143]	Images	Semantic Compression	Multi-scale GAN	Multi-scale latent semantics	TCAV, Disentanglement Metrics
[144]	Image/Video	Semantic Source Encoding	DNN	Frame-level semantic saliency	Temporal Attribution
[145]	Images	Semantic Reconstruction	CNN	Evidence for semantic similarity	Example-based Explanations
[146]	Images	Semantic Source Encoding	ResNet56	Class-discriminative features	Grad-CAM, IG
[147]	Text	Semantic Source Encoding	Knowledge Graphs	Entity and relation importance	Graph-based Attribution
[148]	Text	Semantic Source Encoding	Graph Transformer	Semantic node influence	Attention Visualization
[149]	Audio/Speech	Semantic Reconstruction	CNN (SE-ResNet)	Time-frequency contribution	Attribution, Saliency Maps
[151]	Text	Semantic Source Encoding	Transformer	Token importance	Integrated Gradients, Attention
[150]	Source Signal	Semantic-aware Channel Adaptation	Reinforcement Learning	State-action importance	Policy Explainability (SHAP for RL)
[152]	Text	Semantic Compression	Transformer	Latent semantic bottleneck	Latent Traversal, TCAV
[153]	Text	Semantic Reconstruction	LSTM-RNN	Word-level contribution	Attribution
[154]	Multimodal	Multimodal Semantic Fusion	GAN + ResNet	Cross-modal concept alignment	Concept-based XAI
[155]	Images	Semantic-aware Channel Adaptation	RL-ASC	Policy behavior	Surrogate Policy Trees
[156]	Audio	Semantic Source Encoding	CNN	Spectral importance	Saliency Maps
[157]	Text	Semantic Compression	Transformer	Bottleneck token importance	Feature Importance
[158]	UAV Actions	Semantic-aware Control	MADRL	Agent decision rationale	Policy Summaries
[159]	Images	Semantic Source Encoding	ResNet	Attention maps	Attention Visualization
[160]	Text	Semantic Reconstruction	Transformer	Token attribution	SHAP
[161]	Images	Semantic Reconstruction	Transformer	Semantic segments	Semantic Segmentation + Attribution

The study presented in [141] introduces an innovative end-to-end image transmission system that leverages semantic communication and integrates AI technologies with 6G communication networks. This system, consisting of a transmitter (encoder) and a receiver (decoder), communicates over a physical channel. The transmitter performs a semantic segmentation task on the image to extract its semantic meaning, transmitting the segmentation map instead of the original image. At the receiving end, a pre-trained GAN network reconstructs a realistic image from the received segmentation map. Using the COCO-Stuff dataset for training, both the transmitter and receiver share a common knowledge base. The research also examines the impact of physical channel distortions and quantization noise on the transmission of multimedia content via semantic communication. Experimental results highlight the system's efficiency, demonstrating a compression ratio of approximately 20 compared to traditional image transmission methods.

In their work [130], the authors propose a novel communication model called semantics-native communication (SNC), which introduces a stochastic approach to semantic communication. This SNC system operates in a point-to-point communication scenario where a speaker aims to convey information about an entity. Semantic concepts are extracted and translated into symbols, which are then sent to a target listener through a Shannon communication channel. A key innovation of the SNC framework is its incorporation of contextual reasoning, inspired by human communication, where the speaker considers the listener's context before transmitting information. In this framework, both the speaker and listener engage in iterative communication with a virtual agent that possesses the listener's state, using these agent states as additional information. Theoretical analysis confirms the convergence of contextual reasoning, ensuring minimal bit length while maintaining the desired level of reliability. Simulation results demonstrate the effectiveness of the contextual reasoning-based SNC, showing a significant reduction in bit lengths and offering a robust solution even when the agent states are imperfect.

The study presented by the authors in [142] introduces a multi-task semantic communication system that leverages joint source-channel coding (JSCC) to transmit semantic symbols for various AI tasks. The system is designed to efficiently extract and transmit common semantics shared among multiple tasks, thereby minimizing the overall amount of information transmitted. By framing the multi-task communication as a rate-distortion problem, the system uses an advanced rate-distortion theory to find the best balance between the rate of extracted seman-

tics and the distortion experienced across different tasks. A notable feature of this system is the dynamic rate adjustment module, which optimizes performance based on current channel conditions. Experimental results demonstrate the system's effectiveness, showing a remarkable 7.6 percent improvement in multi-task performance and a 40 percent reduction in transmission overhead compared to single-task systems. This efficiency is achieved through the dynamic rate adjustment and the compact extraction of semantics. The system's performance is evaluated using metrics such as semantic fidelity loss, multi-task performance (MTL), and transmission overhead reduction, with quantification through mean intersection over union (mIoU) and root mean square error (RMSE). These results highlight the system's capability to transmit semantics across a range of AI tasks efficiently.

The authors in [143] present a groundbreaking communication paradigm that revolves around AI models, focusing on model propagation. This innovative approach uses semantic slice models (SeSM) to minimize communication overhead and allow for adaptable model configurations based on different requirements. The system achieves highly efficient semantic communication by integrating AI models into both the transmitter and receiver at the physical layer. The system model is tailored to specific sources, scenes, and tasks, with AI models adapting to these contextual constraints. One of the critical aspects explored in the paper is the balance between the longevity of AI models and how often they need to be propagated. Although the goal is to optimize end-to-end AI models for semantic communication, a significant challenge is effectively segmenting these models to meet specific objectives. The proposed system includes an encoder, a generator, enhancement slice models, and channel coding slice models, each with unique network structures and processes. The system also takes into account computational complexity and channel conditions, leading to the use of various channel coding schemes. The paper highlights the synergy of AI models, semantic communication principles, and optimization techniques to enable effective communication and model propagation.

The SparseSBC system presented in [144] introduces a novel approach for visual transmission using semantic communication. This system employs two distinct Deep Neural Network (DNN)-based models: one at the transmitter and one at the receiver. The innovation lies in an alternating learning process where encoding and decoding are continuously refined. The DNN-based transmitter generates a sparse set of bits by utilizing "semantic bases" through a binary quantization module. SparseSBC uses a "self-critic" training scheme to maintain

stable training. This system stands out for its impressive transmission efficiency and effectiveness across various channel conditions, outperforming traditional SemCom solutions. The evaluation of SparseSBC is conducted using the Cifar-10 dataset, which includes 60,000 RGB images, as well as a recorded video clip. The performance metrics include the number of transmitted bits, peak signal-to-noise ratio (PSNR), and structural similarity index measure (SSIM). Simulation results consistently show that SparseSBC achieves superior transmission performance compared to other SemCom solutions like JSCC and MLSC.

The paper [145] presents an innovative blockchain framework for secure semantic communication in the Metaverse, tailored for AI-Generated Content (AIGC) services in virtual transportation networks. Virtual Service Providers (VSPs) and edge devices interact within this framework, where edge devices provide semantic data for diverse services, including AIGC. To address security issues with potentially malicious semantic data, the paper proposes a defense scheme using blockchain and zero-knowledge proofs. This scheme distinguishes between authentic and adversarial semantic data, verifying transformations' legitimacy. The verification process occurs on the blockchain, allowing decentralized querying of results by edge devices and VSPs, fostering trust. The simulation results demonstrate the defense method's effectiveness, achieving up to a 30 percent reduction in semantic similarity between adversarial and authentic data compared to the attack scheme. The blockchain-aided semantic communication enhances AIGC service security, preventing malicious injections. The framework supports seamless interactions between physical and virtual domains, enabling VSPs to offer diverse services through advanced techniques like AIGC, while optimizing data transmission efficiency.

The authors in [146], introduces a novel joint device edge digital semantic communication system featuring trained non-linear quantization. The system is designed to efficiently transmit semantic features, addressing challenges encountered in analog semantic communication. It incorporates a unique non-linear quantization module and a sparse scaling vector to reduce feature dimensionality. Experimental findings reveal that the proposed non-linear quantization outperforms linear quantization, and the digital system surpasses analog semantic communication. Evaluation is conducted using the CIFAR-10 image dataset, demonstrating that the proposed system experiences less than 1 percent accuracy degradation due to quantization at an SNR of 6dB. The newly proposed quantization method exhibits approximately a 4 percent enhancement in accuracy compared to the linear quantization approach. Overall, these results underscore the superior ability of the proposed digital system to effectively represent semantic features.

The proposed system model in [147] introduces a semantic communication framework tailored for transmitting textual data. Leveraging a knowledge graph (KG) composed of semantic triples, the base station (BS) extracts meaningful information from the text, subsequently transmitting this semantic content to users. A graph-to-text generation model at the user end facilitates the recovery of the original text, and the framework's performance is evaluated using the semantic similarity metric (MSS), capturing the accuracy and completeness of the recovered text. To enhance the total MSS of all recovered texts while meeting transmission delay requirements, the system model optimizes resource block (RB) allocation and determines partial semantic information. Operating in a continuous state space, the model incorporates a policy dictating the probability of the BS, acting as an agent, making decisions on RB allocation and selecting partial semantic information for each user. Simulations in a circular network, comparisons with five baselines, and illustrations of the correlation between semantic triples and tokens underscore the framework's efficiency in reducing data size while preserving meaningful content.

The paper [148] presents a semantic communication framework tailored for wireless networks, where a base station (BS) extracts semantic information from textual data and transmits it to users using a knowledge graph (KG) representation. The semantic information, or-

ganized as semantic triples, is employed by each user to recover the original text through a graph-to-text generation model. The system's effectiveness is quantified using the semantic similarity (MSS) metric, encompassing both semantic accuracy and completeness. Despite wireless resource constraints, the BS optimally transmits partial semantic information to meet transmission delay limits. The optimization problem of resource allocation and semantic information selection is addressed using a policy gradient-based reinforcement learning algorithm integrated with an attention network, showcasing significant reductions in data transmission size and a two-fold enhancement in MSS compared to traditional communication networks. In comparative evaluations, the proposed framework surpasses two baselines, a conventional policy gradient algorithm for resource block allocation and a symbols-based wireless communication scheme, exhibiting superior data reduction and MSS performance. Simulations conducted in a circular network with one BS and uniformly distributed users confirm the framework's efficacy. By prioritizing the transmission of crucial semantic information, the proposed framework demonstrates a streamlined approach to recovering the primary meaning of the original text while minimizing data transmission requirements.

In [149], a semantic communication system for speech signals is presented which consists of a transmitter and a receiver, where the transmitter incorporates a semantic encoder and a channel encoder, implemented through independent neural networks (NN) denoted as 'a' and 'b', respectively. Utilizing these NN parameters, the transmitter generates an encoded symbol sequence 'x'. The receiver, employing a deep learning approach, reconstructs the raw speech signals from the encoded symbol sequence. Emphasizing semantic-level error minimization over bit or symbol levels, the training process incorporates an attention mechanism based on a squeeze-and-excitation (SE) network to identify critical speech information. Crucially, the system is designed for adaptability to dynamic channel conditions without necessitating retraining. The proposed DeepSC-S semantic communication system showcases enhanced transmission efficiency and accuracy in recovering speech signals at the semantic level. Demonstrating superiority over traditional communication methods, DeepSC-S excels in speech signal metrics such as signal-to-distortion ratio (SDR) and perceptual evaluation of speech distortion. Notably, it exhibits robustness in low signal-to-noise ratio (SNR) scenarios, adapting effectively to dynamic channel variations without retraining. The system's design, incorporating a deep learning approach with an attention mechanism based on a squeeze-and-excitation (SE) network, ensures the identification of crucial speech information during training, contributing to its overall effectiveness in diverse communication scenarios.

The paper [151], introduces a novel system model tailored for wireless resource management in the context of an Intelligent Semantic Communication Network (ISC-HetNet). This model leverages an auxiliary knowledge base (KB) alongside AI models to effectively encode semantic content, enhancing the efficiency of communication. The primary goal is to maximize the system throughput in messages (STM), addressing challenges related to user association (UA) and bandwidth allocation (BA) through a joint optimization approach that considers KB matching and wireless bandwidth constraints. The proposed two-stage solution, integrating stochastic programming and a heuristic algorithm, demonstrates superior performance in improving system throughput in message (STM) compared to baseline algorithms. Implemented within an ISC-HetNet, the proposed system achieves heightened STM by integrating an auxiliary KB and employing AI models for semantic content encoding. The joint optimization of user association (UA) and bandwidth allocation (BA) plays a crucial role in maximizing STM while considering KB matching and wireless bandwidth constraints. Numerical results validate the efficacy of the two-stage solution, consistently showcasing its superiority and reliability in enhancing the system's STM performance when compared to established baseline algorithms.

The authors in [150], introduces a pioneering communication architecture, Implicit Semantic-Aware Communication (iSAC), designed to

comprehensively address both explicit and implicit semantic nuances in communication. It innovatively employs a graph-inspired representation encompassing explicit semantics (objects and relations), implicit semantics (inferred information behind entities), and reasoning mechanisms (user's reasoning rule and inference policy). To enhance the transmission and decoding of semantic information, the paper proposes a projection-based semantic encoding solution. This solution efficiently converts explicit semantics into a concise, low-dimensional semantic constellation space. Furthermore, the paper introduces a cutting-edge solution, Generative Imitation-based Reasoning Mechanism Learning (G-RML), to automate the encoding, transportation, and decoding/interpretation of implicit semantic meaning at the destination user. The proposed system undergoes rigorous evaluation through extensive simulations using real-world human knowledge datasets. The evaluation metrics include the semantic error rate, Signal-to-Noise Ratio (SNR), runtime efficiency, and implicit semantic inference accuracy, providing a comprehensive assessment of the system's performance. The paper's significant contributions lie in the development of the iSAC architecture, the introduction of novel solutions for semantic encoding and reasoning, and the thorough evaluation of the system's capabilities through simulations.

A DeepSC model is presented in [152], which is an intelligent end-to-end communication system, operates on two key levels: semantic and transmission. The semantic level focuses on processing information for encoding and decoding, while the transmission level ensures accurate exchange over the medium. DeepSC aims to maximize system capacity, emphasizing sentence meaning recovery over traditional bit or symbol error concerns. Utilizing transfer learning, it adapts to diverse communication environments, demonstrating superior performance in low signal-to-noise ratio scenarios and increased resilience to channel variation. The comparisons with traditional systems highlight DeepSC's robustness, particularly in low SNR regimes, across varied channel conditions. Introducing the novel metric "sentence similarity," the system precisely measures semantic communication performance, justifying DeepSC's efficacy in meaning recovery and semantic error reduction. An analysis of loss value, mutual information, BLEU score, and SNR underscores DeepSC's performance, revealing the impact of training parameters on mutual information and BLEU score. While the sources lack explicit overall results, they provide valuable insights into DeepSC's enhanced performance and the metrics used for evaluation.

The system proposed in [153], aims to ensure the semantic consistency of Aviation Radiotelephony Communication (ARC) between pilots and air traffic controllers. It utilizes both Recurrent Neural Network (RNN) and Long Short-Term Memory Recurrent Neural Network (LSTM-RNN) architectures, training on a corpus derived from recorded communications. Sentences are annotated as either consistent or non-consistent, and words are represented as one-hot vectors using an aviation-specific glossary. The models are trained through supervised learning, and semantic consistency is verified using cosine similarity and a sigmoid layer. Optimization during training employs the Nesterov method to address gradient problems. The results indicate that LSTM-RNN outperforms the standard RNN, achieving an average test accuracy of 67.02 percent compared to 61.44 percent. Additionally, the smaller Mean Squared Error (MSE) for LSTM-RNN suggests more stable performance across multiple tests. The findings support the conclusion that LSTM-RNN is a practical solution for improving the semantic consistency of ARC, thereby enhancing aircraft flight safety by reducing communication errors between pilots and air traffic controllers.

The paper [154], introduces a comprehensive system model for cross-modal retrieval and semantic representation learning. It incorporates generative parts for image and text representation, classification parts for multi-label classification, and an inter-modal discriminative network to bridge the modality gap. Adversarial learning is employed to iteratively train generative and discriminative models, aiming to generate common semantic representations and distinguish differences between modalities. The joint optimization of the generative adversar-

ial network and classification network enhances cross-modal retrieval. The proposed model demonstrates superior performance on widely used datasets, achieving indistinguishable and close representations for data with similar semantics through adversarial learning. The inter-modal discriminative network effectively contributes to overcoming the modality gap by distinguishing between image and text modalities.

[155] introduces an Adaptive Image Compression System (AICS) based on a Reinforcement Learning (RL) framework incorporating Semantic Segmentation (SS). The RL agent, responsible for analyzing the image state, develops a policy for adaptive quantization, while the Semantic Decoder (SD) reconstructs the image based on the received bitstream and the chosen quantization level. This RL framework encompasses a State Space (SS) representing all possible information observed by the agent, an Action Space (AS) defining the set of quantization levels, and a Policy (P) that maps states to a distribution of actions, facilitating effective learning. The evaluation of the system's performance employs metrics such as Boundary Displacement mean Intersection over Union (BD-mIoU) and Boundary Displacement rate (BD-rate), with a specific focus on Semantic Segmentation Tasks (SST). BD-mIoU quantifies segmentation quality by assessing the intersection over union between predicted and ground truth segmentation, while BD-rate measures the system's segmentation performance relative to established benchmarks. These metrics play a crucial role in assessing the effectiveness of the system in accurately segmenting objects and capturing semantic information in the compressed images. The primary goal is to achieve higher BD-mIoU values and negative BD-rate values, indicating enhanced segmentation performance and improvements over existing baselines. In essence, the proposed AICS leverages RL and SS to dynamically adapt image compression, aiming to achieve superior segmentation quality compared to established benchmarks, and the evaluation metrics provide a comprehensive assessment of the system's efficacy, emphasizing the importance of achieving higher BD-mIoU values and negative BD-rate values as key indicators of improved segmentation performance.

The system model in [156] is built on a wav2vec-based autoencoder architecture, employing convolutional neural networks (CNNs) in its multiple layers. The initialization process includes setting up the architecture, local models, and the global model with random parameters. The encoder and decoder of the autoencoder collaboratively update parameters simultaneously through stochastic gradient descent (SGD). The system involves wireless transmission of audio data, where the transmitted data is used to calculate Local Normalized Root Mean Square Error (LNRMSSE). The server periodically collects and aggregates local models, broadcasting the updated global model back to each device. Performance evaluation is conducted using metrics such as Mean Squared Error (MSE), Bit Error Rate (BER), and data transmission volume, with testing performed on speech data from the Librispeech dataset. The semantic method significantly outperforms the traditional approach, reducing transmission MSE by nearly 100 times, indicating enhanced accuracy and noise immunity. Additionally, the semantic method exhibits more stable MSE variation, and as the Signal-to-Noise Ratio (SNR) increases, communication errors decrease. The proposed autoencoder's performance is further assessed using validation loss during training epochs. The study reveals that the MSE of the semantic method shows a flatter variation compared to the traditional method, suggesting improved stability in transmission. Higher SNR correlates with reduced communication errors due to milder noise, whereas the transmission MSE of the traditional method remains constant beyond an SNR of 14 dB due to coding limitations. Overall, the research demonstrates the efficacy of the wav2vec-based autoencoder in enhancing audio transmission accuracy over wireless networks, particularly in scenarios with varying SNR levels.

The paper [157] introduces a novel semantic communication system leveraging the Universal Transformer model, which concurrently encodes semantic source and performs channel coding over a noisy channel. The system architecture includes a transmitter utilizing the Universal Transformer encoder and a receiver with a channel decoder

responsible for symbol detection and a semantic decoder for text recovery. Comparative evaluations against existing algorithms and traditional joint coding approaches are conducted under diverse channel conditions, such as AWGN and Rayleigh fading channels. The results highlight the superior end-to-end performance of the proposed system in semantic communication. The adaptive Universal Transformer's incorporation introduces enhanced flexibility for accommodating varied semantic information in transmitted sentences. While acknowledging potential SER error floors, the system exhibits promising effectiveness in achieving semantic communication over noisy channels, particularly within specific Signal-to-Noise Ratio thresholds.

In [158], the authors introduce a pioneering multiagent deep reinforcement learning (MADRL) framework named Graph Attention Exchange Network (GAXNet) for optimizing air-to-ground ultra-reliable and low-latency communication (URLLC) using unmanned aerial vehicles (UAVs). GAXNet utilizes a graph attention mechanism and semantic communication to enhance cooperation among UAV agents. Through the exchange of semantic weights based on complete and partial information, attention mismatches are reduced, improving the reliability of the air-to-ground network in terms of latency and error rate. The system model, incorporating principles of Centralized Training and Decentralized Execution (CTDE) MADRL, manages real-time control of UAVs, ensuring collision avoidance. Evaluation in a simulated environment involves rewards during training, network reliability, joint-action value function, utility function, and attention mismatch reduction, providing a comprehensive analysis of GAXNet's performance.

The authors in [159], propose a novel task-oriented semantic communication framework (X-SED) that prioritizes transmitting only task-relevant semantic information rather than reconstructing the full message. This approach significantly reduces bandwidth usage while maintaining functional performance for downstream tasks. A key innovation in their work is the integration of explainability into the semantic encoder-decoder process through attention visualization and gradient-based saliency techniques, which help identify which input features most influence the transmitted output. These interpretability tools offer valuable insight into the model's decision making process, enabling users and developers to understand, validate, and trust how semantic content is selected and compressed. The study thus marks a crucial advancement toward making semantic communication systems more transparent, efficient, and trustworthy especially in safety critical and bandwidth constrained applications.

Shi et al. in [160], propose XSC-Net, an explainable semantic communication framework tailored for text transmission tasks. The model aims to enhance semantic understanding and efficiency while ensuring that the decision-making process remains interpretable. By integrating SHAP (SHapley Additive exPlanations), the authors provide a transparent view into the importance of each word or semantic unit during encoding, enabling deeper insights into the system's behavior and facilitating trust in AI-driven communications. Experimental evaluations on BLEU scores show that XSC-Net outperforms traditional baselines while maintaining explainability. The inclusion of SHAP not only aids in debugging and model understanding but also promotes user confidence, which is essential for real world deployment of semantic communication systems.

In [161] the authors introduce a trustworthy Image Semantic Communication (ISC) framework that emphasizes explainability and controllability for efficient image transmission. The proposed system converts images into explainable semantic representations comprising semantic text and segmentation maps using pre-trained models like CoCa, YOLOv8, and FastSAM. These interpretable semantics are then processed via a decoupled transeiver architecture and reconstructed at the receiver using a generative adversarial module called ES-IRM, which incorporates CLIP and GALIP models. A multi-rate semantic-aware transmission protocol is proposed, enabling task-specific, feedback-driven communication with minimal redundancy. Explainability is achieved by explicitly encoding and leveraging semantics, allowing transparent,

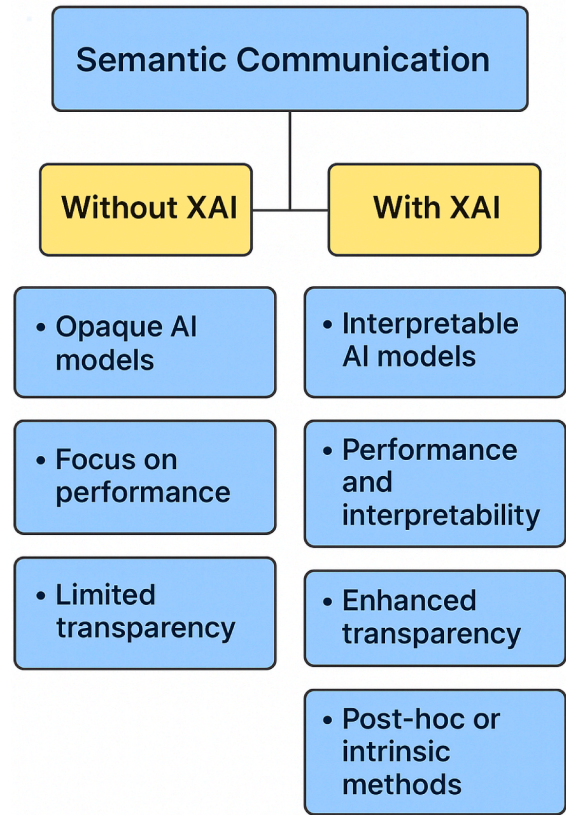


Fig. 9. Simple representation of Black-box AI and explainable Ai model.

task-relevant decisions. Evaluation on the COCO dataset demonstrates superior performance in BLEU, LPIPS, and FID metrics, showing the framework's effectiveness in reducing bandwidth requirements while preserving semantic fidelity.

In conclusion, this section has examined a wide array of cutting-edge developments in semantic communication, highlighting diverse methodologies aimed at enhancing the efficiency, adaptability, and reliability of AI-enabled communication systems. From multi-task joint source-channel coding to semantic slice models and generative adversarial networks for image reconstruction, the surveyed works offer innovative strategies to address key challenges such as bandwidth reduction, semantic fidelity, and system scalability. These advancements collectively underscore the transformative potential of semantic communication in shaping next-generation wireless systems by enabling meaning-centric transmission with reduced overhead and improved performance across varied data modalities.

A notable observation from our literature analysis is the emerging integration of explainable AI (XAI) techniques within semantic communication frameworks. While traditional approaches largely focus on performance and task accuracy using opaque models, recent studies emphasize the importance of interpretability and transparency to enhance trust, debuggability, and user understanding especially in mission-critical and real-time communication scenarios. Fig. 9 provides a conceptual comparison between semantic communication systems with and without XAI, derived from the recent contributions discussed. It illustrates the growing trend toward systems that balance high performance with explainability, using post-hoc or intrinsic methods to bridge the gap between human understanding and AI decision-making. As semantic communication continues to evolve, integrating such trustworthy AI features will be essential for building intelligent, transparent, and robust communication infrastructures suited for complex real-world applications.

Table 9
Summary of explainable AI techniques for semantic communication.

XAI Technique	Applicable Modality	SemCom Layer	Explanation Principle	Runtime & Memory Overhead	Key Limitations / Failure Modes
Rule-Based Explanations	Semantic decision layer (text, symbolic, multimodal)	(text, symbolic, multimodal)	Surrogate model $g(s) \approx f_\theta(s)$ expressed as logical rules	Low-moderate; scales with number of rules $O(Kd)$; low memory	Oversimplifies non-linear semantics; limited fidelity under distribution shift
Feature Importance (SHAP, LIME)	Semantic representation and decision layers	Semantic representation and decision layers	Decomposition $f_\theta(s) \approx \sum_i \phi_i$ via perturbation or Shapley values	Moderate-high; sampling-based; memory for perturbations	Sensitive to feature correlation; instability (LIME); high cost (SHAP)
Visualization (LRP, Saliency)	Encoder/decoder layers (image, waveform, embeddings)	Encoder/decoder layers (image, waveform, embeddings)	Relevance propagation R_i via backward redistribution	Comparable to backpropagation; activation storage required	Noisy relevance maps; resolution loss; architecture sensitivity
Knowledge Graphs & Ontologies	Semantic reasoning and decision layers	Semantic reasoning and decision layers	Inference paths in graph $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathcal{R})$	Query-dependent; scalable on servers; higher memory footprint	Incomplete knowledge; maintenance cost; symbolic-neural mismatch
Rule Extraction	Semantic control and decision layers	Semantic control and decision layers	Explicit rule set $\mathcal{R}$ approximating f_θ	Low-moderate; depends on rule learning strategy	Loss of fidelity; rule explosion; poor scalability in high dimensions
Counterfactual Explanations	Semantic representation and decision layers	Semantic representation and decision layers	Optimization: $\min d(s, s')$ s.t. $f_\theta(s') \neq f_\theta(s)$	Moderate-high; iterative search or generative sampling	Unrealistic counterfactuals; spurious correlations; metric sensitivity
Explainable Reinforcement Learning	Semantic control and adaptation layer	Semantic control and adaptation layer	Policy explanation in MDP $(\mathcal{S}, \mathcal{A}, \mathcal{R})$	Moderate; gradients, trajectory storage, surrogate models	Instability across training; difficult long-term credit assignment
Natural Language Generation (NLG)	Explanation and interaction layer	Explanation and interaction layer	Text generator $g_\phi(s, E) \rightarrow \text{explanation}$	Model-size dependent; token generation cost	Hallucination; faithfulness gap; latency if verbose

4. XAI techniques for semantic communication

As semantic communication systems continue to mature and incorporate higher levels of intelligence and contextual reasoning, the question of how their decisions can be understood and trusted becomes increasingly important. Earlier sections discussed the motivation for explainable artificial intelligence (XAI) in semantic communication and outlined its potential benefits. Building on that discussion, this section turns to the practical techniques that enable interpretability within semantic communication pipelines. We focus on methods that explain how semantic encoders, decoders, and intermediate control components prioritize, transform, and transmit meaning. By examining how different XAI approaches capture semantic relevance, feature contribution, and decision logic, this section clarifies their role in improving transparency, adaptability, and user confidence in AI-enabled communication systems. Fig. 10 offers a visual overview of representative XAI techniques applicable to semantic communication, while Table 9 provides a comparative summary of their formal principles, computational overhead, and limitations, which are discussed in detail throughout this section.

While model-agnostic attribution methods such as LIME and SHAP are primarily designed to explain how input tokens, regions, or signal segments influence semantic encoding and decoding, they alone cannot fully capture the structure of the *intermediate latent manifold* employed in semantic compression. Consequently, XAI-enabled semantic communication additionally relies on *concept-based and latent-space analysis methods*, such as Testing with Concept Activation Vectors (TCAV), latent traversal with counterfactual decoding, and disentanglement analysis, to characterize how high-level semantic factors are organized within the compressed representation. These approaches enable the association of human-interpretable concepts with directions or subspaces in the latent code and provide insight into the semantic meaning of latent dimensions, thereby complementing input-output attribution techniques and supporting deeper interpretability of semantic encoders.

Below, we discuss in detail representative XAI techniques for semantic communication and highlight their applicability, advantages, and limitations across different layers of the SemCom pipeline.

1. Rule-Based Explanations. Rule-based explanation techniques aim to make AI decisions transparent by expressing them through explicit logical rules of the form *if-then*. In semantic communication systems, such methods are naturally aligned with the semantic decision layer, where high-level inference outcomes depend on interpretable se-

mantic attributes rather than raw signal representations. Let $f_\theta : S \rightarrow \mathcal{Y}$ denote the semantic decision function learned by a neural semantic encoder-decoder pipeline, where S represents the semantic feature space and $\mathcal{Y}$ denotes the task output. Rule-based explanations approximate f_θ using an interpretable surrogate function $g : S \rightarrow \mathcal{Y}$, which is expressed as a finite set of logical rules $\mathcal{R} = \{r_1, r_2, \dots, r_K\}$, with each rule encoding a conjunction of semantic conditions leading to a decision outcome [94].

In practice, the surrogate model g is commonly derived from decision trees or logical rule sets that locally or globally approximate the behavior of the underlying black-box semantic model. For a given semantic instance $s \in S$, the explanation identifies the subset of activated rules $\mathcal{R}(s) \subseteq \mathcal{R}$, thereby decomposing the decision into a small number of human-readable semantic conditions [162,163]. This form of explanation is particularly effective in semantic communication scenarios where decisions are driven by interpretable contextual cues, such as user intent, object presence, or domain-specific attributes.

A representative framework in this category is *FACETS* (Feature-based Explanation for Counterfactuals and Explanations via Trees), which generates rule-based and counterfactual explanations by identifying salient semantic features that influence model predictions [164]. Within semantic communication pipelines, FACETS can be applied either to intermediate semantic representations or to final decision outputs by learning a rule-based approximation of f_θ over the semantic feature space. The resulting facet-wise decomposition ranks semantic elements according to their influence, enabling both instance-level and model-level interpretability. From a computational standpoint, rule extraction typically incurs low to moderate overhead, with time complexity on the order of $O(K \cdot d)$ for K extracted rules over d semantic features, and memory requirements dominated by storing the rule set itself. As a result, rule-based explanations are well suited for deployment on resource-constrained edge devices.

Use Case Example: Consider an intelligent healthcare system in which patient symptoms are transmitted via semantic communication to a diagnostic module. Let $s = [s_1, s_2, \dots, s_d]$ denote the semantic feature vector encoding clinical attributes such as fever intensity, cough duration, and medical history. FACETS can generate interpretable rules such as “if $s_{\text{cough}} > 3 \text{ days}$ and $s_{\text{fever}} > 101^\circ\text{F}$, then diagnosis = pneumonia,” which explicitly reveal the semantic basis of the predicted outcome. Such explanations provide clinicians with transparent decision traces that can be evaluated against medical expertise, thereby increasing trust in safety-critical, real-time communication scenarios.

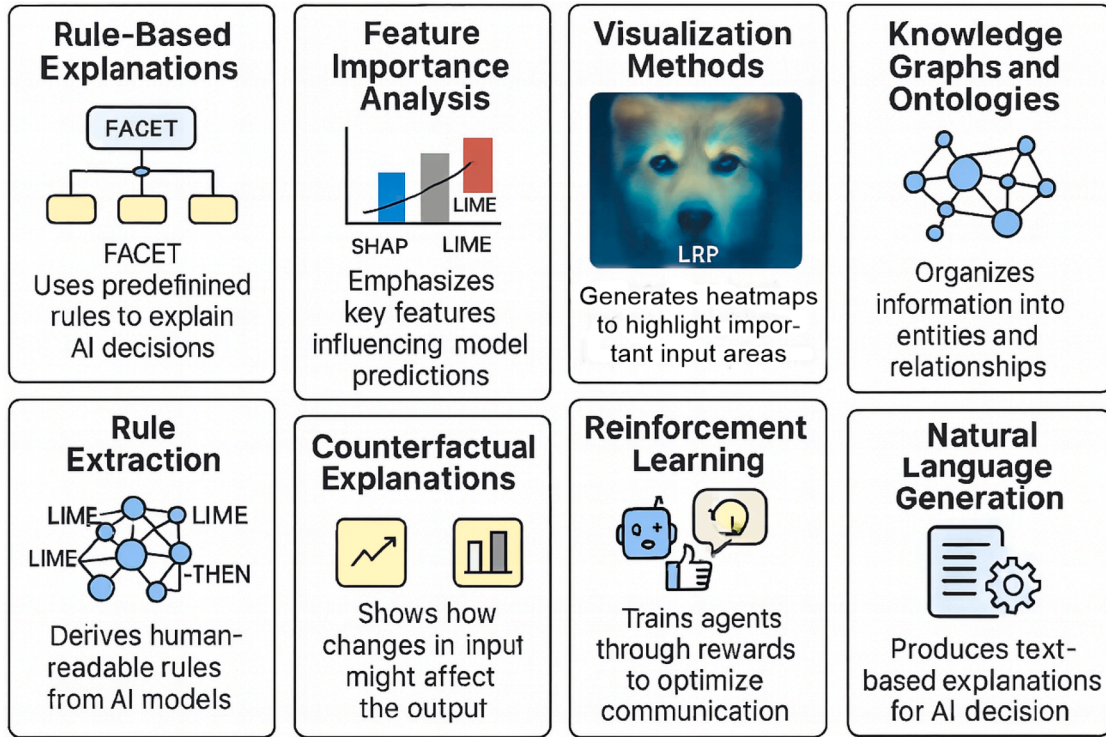


Fig. 10. State of the art XAI techniques for semantic communication.

Despite their interpretability advantages, rule-based explanations also exhibit limitations. The surrogate function g may fail to faithfully capture highly non-linear interactions present in deep semantic models, leading to reduced explanation fidelity. Moreover, rule sets may oversimplify complex semantic dependencies, become unstable under distribution shifts, or scale poorly as the dimensionality of the semantic feature space increases. Consequently, rule-based explanations are best viewed as complementary mechanisms that provide lightweight and intuitive insights alongside more expressive explanation techniques in XAI-enabled semantic communication systems.

2. Feature Importance Analysis. Feature importance analysis aims to identify and quantify the contribution of individual input features to a model's prediction, thereby offering insight into how decisions are formed. In the context of semantic communication systems, feature importance methods are particularly valuable at the semantic representation and decision layers, where understanding the influence of linguistic tokens, visual attributes, or contextual cues is essential for preserving meaning during encoding and decoding. By highlighting which semantic components drive transmission or reconstruction outcomes, these methods enhance transparency and help verify semantic fidelity under compression and channel constraints [165].

Let $f_\theta : S \rightarrow \mathcal{Y}$ denote a trained semantic communication model that maps semantic features $s \in S$ to an output $y \in \mathcal{Y}$. Feature importance analysis seeks to decompose the prediction $f_\theta(s)$ into contributions associated with individual semantic features s_i , such that

$$f_\theta(s) \approx \sum_{i=1}^d \phi_i \quad (24)$$

where ϕ_i represents the importance attributed to the i -th semantic feature. This decomposition can be performed either globally, capturing average feature influence across the dataset, or locally, explaining individual semantic decisions. Both perspectives are relevant in SemCom systems, where global trends inform model design and local explanations support trust and debugging.

SHAP (SHapley Additive exPlanations) is a game-theoretic approach that assigns feature attributions based on Shapley values, ensuring properties such as local accuracy, consistency, and missingness [99]. SHAP is well suited for semantic communication models operating on high-dimensional semantic representations, as it provides both global and instance-level explanations grounded in a rigorous mathematical framework. Given a semantic feature set S , SHAP computes ϕ_i as the expected marginal contribution of feature s_i over all feature coalitions. However, this theoretical rigor comes at a computational cost: practical SHAP implementations typically rely on sampling or model-specific approximations, resulting in runtime complexity that scales with the number of perturbation samples and requiring additional memory to store intermediate evaluations. Moreover, SHAP explanations can become unreliable in the presence of highly correlated semantic features, a common scenario in learned embeddings.

LIME (Local Interpretable Model-Agnostic Explanations) provides local feature importance by fitting an interpretable surrogate model around a specific prediction [166,167]. For a given semantic instance s , LIME samples perturbed versions of s and learns a simple model g that approximates f_θ in the local neighborhood. This makes LIME particularly attractive for explaining instance specific semantic decisions, such as why certain words, symbols, or visual regions were prioritized during semantic encoding. From a computational standpoint, LIME incurs overhead proportional to the number of perturbations used to train the local surrogate, which may limit scalability for high dimensional semantic inputs. In addition, LIME explanations are sensitive to the choice of perturbation distribution and may lack stability across similar semantic instances.

Use Case Example: In an autonomous vehicle communication scenario where vehicles exchange compressed semantic representations of traffic scenes, feature importance analysis can be used to explain object level or region level decisions. If a semantic decoder incorrectly classifies a road sign as a pedestrian, LIME can highlight the specific visual regions or semantic attributes that contributed most to the misclassification, while SHAP can reveal whether certain semantic features systematically bias

hazard detection across multiple scenes. Such insights support model debugging and accountability in real-time, safety critical communication environments.

Overall, SHAP and LIME provide complementary perspectives on feature importance in semantic communication systems. SHAP offers theoretically grounded and consistent attributions suitable for global analysis, whereas LIME provides flexible, instance level explanations that are easy to interpret. At the same time, both methods exhibit limitations related to computational overhead, feature correlation, and explanation stability, which must be carefully considered when deploying them in resource constrained or latency sensitive SemCom applications.

3. Visualization Methods. Visualization based explanation techniques enhance interpretability by providing intuitive, human-readable insights into how semantic communication models process and transmit information. These methods are particularly relevant for semantic encoders and decoders that operate on structured or spatial representations, such as images, spectrograms, or token level embeddings. By visually highlighting the regions, features, or semantic units that most strongly influence a model's output, visualization methods help elucidate the internal decision pathways of AI-driven communication systems [168].

Formally, let $f_\theta : \mathcal{X} \rightarrow \mathcal{Y}$ denote a semantic encoder-decoder model, where $\mathcal{X}$ represents the input space (e.g., images, token sequences, or waveform features) and $\mathcal{Y}$ denotes the semantic output. Visualization methods aim to compute a relevance map $R \in \mathbb{R}^{|\mathcal{X}|}$ such that each element R_i reflects the contribution of input component x_i to the model output $f_\theta(x)$. This formulation makes visualization techniques well suited to modalities with inherent spatial or sequential structure, where semantic importance can be meaningfully localized.

A representative technique in this category is *Layer-wise Relevance Propagation (LRP)*, which assigns relevance scores to neurons and input features by redistributing the prediction score backward through the network layers [117]. Given an output activation $f_\theta(x)$, LRP propagates relevance according to conservation rules that ensure the total relevance is preserved across layers. In semantic communication systems, LRP is particularly effective for analyzing how semantic information is encoded, compressed, and reconstructed across deep neural layers, thereby offering insight into both local feature relevance and global semantic behavior [118,169,170].

From a computational perspective, visualization methods such as LRP typically require one or more backward passes through the network, resulting in runtime complexity comparable to standard gradient-based inference. Memory overhead is dominated by storing intermediate activations during backpropagation, which makes such methods feasible for edge deployment when applied selectively or at reduced resolution. However, visualization techniques are not without limitations. They may produce noisy or diffuse relevance maps, struggle to capture long range dependencies in deep architectures, and can be sensitive to architectural choices or normalization strategies. In addition, visualization quality may degrade for highly compressed semantic representations, where spatial or feature-level resolution is reduced.

Use Case Example: Consider a semantic image transmission system in which a deep encoder-decoder model compresses visual content prior to wireless transmission. Applying LRP to the semantic encoder enables visualization of the image regions (e.g., faces, vehicles, or traffic signs) that contribute most strongly to the semantic representation. Such insights can inform bandwidth allocation strategies by prioritizing semantically salient regions, while also helping identify biases or information loss introduced during compression or transmission.

Overall, visualization methods provide an intuitive and modality-aware means of explaining semantic communication models, particularly for image- and waveform-based systems. When used alongside feature-importance or rule-based explanations, they contribute to a more comprehensive understanding of semantic decision making, while their computational cost and sensitivity to model structure must be carefully

managed in resource-constrained or real time communication environments.

4. Knowledge Graphs and Ontologies. Knowledge graphs (KGs) and ontologies provide structured, machine-readable representations of entities and their relationships, making them well suited for enhancing explainability in semantic communication systems. By encoding domain knowledge in the form of entities, relations, and logical constraints, KGs enable semantic reasoning and inference that can be explicitly traced and interpreted. This property is particularly valuable in semantic communication, where preserving and explaining high-level meaning is more important than transmitting raw data [171,172].

Formally, a knowledge graph can be represented as $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathcal{R})$, where $\mathcal{V}$ denotes the set of semantic entities, $\mathcal{E}$ the set of edges, and $\mathcal{R}$ the set of relation types. Given a semantic message M , semantic encoding can be viewed as a mapping $M \rightarrow S \subseteq \mathcal{V}$, where S represents the activated concepts within the graph. Explainability arises from the ability to trace inference paths $\mathcal{P} = \{(v_i, r_{ij}, v_j)\}$ within $\mathcal{G}$ that justify a semantic decision or transmitted message. This explicit reasoning structure makes KGs particularly appropriate for the semantic reasoning and decision layers of SemCom systems.

From an explainability perspective, KGs and ontologies support transparent decision-making by exposing the logical dependencies and relational paths that lead to a given output. Unlike post-hoc attribution methods, explanations derived from KGs are inherently symbolic and causally interpretable, allowing users to inspect which entities, relations, or rules contributed to the semantic outcome [173,174]. This capability is especially important in regulated or high-stakes domains such as healthcare, finance, and law, where semantic decisions must be auditable and justifiable [175,176].

In terms of computational overhead, KG-based explanation primarily involves graph traversal and reasoning operations, with complexity dependent on graph size and query depth. For typical semantic queries, inference can be performed efficiently using bounded-depth traversal or rule-based reasoning, making KG explanations suitable for cloud or edge-server deployment. However, maintaining and querying large-scale or dynamic knowledge graphs may incur non-trivial memory and update costs, particularly when domain knowledge evolves rapidly or when semantic concepts are weakly defined.

Use Case Example: In an intelligent legal advisory system employing semantic communication, a knowledge graph can encode relationships among legal concepts such as statutes, precedents, and contractual clauses. When a legal query is semantically encoded and transmitted, the system can generate an explanation by tracing the inference path through relevant entities and rules within the KG. This allows end users to understand which legal provisions influenced the transmitted response, thereby validating both the semantic content and the underlying reasoning process.

Despite their advantages, KG- and ontology-based explanations are not without limitations. They rely on the completeness and correctness of the underlying knowledge base, and may fail to capture implicit or learned semantics present in data-driven models. Moreover, constructing and maintaining domain-specific KGs can be labor-intensive, and mismatches between learned semantic embeddings and symbolic representations may reduce explanation fidelity. As a result, knowledge graphs are most effective when used in conjunction with learned semantic models, providing structured, interpretable explanations that complement data-driven semantic communication pipelines.

5. Rule Extraction. Rule extraction techniques aim to distill the behavior of complex AI models into concise, interpretable logical rules of the form *IF-THEN*. Unlike feature attribution methods that assign importance scores to individual inputs, rule extraction focuses on recovering explicit decision logic that approximates the model's behavior. In semantic communication systems, this capability is particularly valuable at the semantic decision and control layers, where understanding how semantic representations lead to specific actions or outputs is essential for trust, verification, and oversight [163].

Formally, let $f_\theta : S \rightarrow \mathcal{Y}$ denote a trained semantic communication model that maps semantic representations $s \in S$ to outputs $y \in \mathcal{Y}$. Rule extraction seeks to construct an interpretable function $g : S \rightarrow \mathcal{Y}$, expressed as a finite rule set $\mathcal{R} = \{r_1, r_2, \dots, r_K\}$, such that $g(s) \approx f_\theta(s)$ within a specified fidelity bound. Each rule r_k encodes a conjunction of semantic conditions that lead to a predicted outcome, enabling direct inspection of the model's decision logic.

Rule extraction methods are commonly categorized into three classes: *decompositional approaches*, which exploit the internal structure of neural networks to extract rules at the level of neurons or layers; *pedagogical approaches*, which treat the model as a black-box and derive rules solely from observed input-output behavior; and *eclectic approaches*, which combine both internal and external information to improve rule fidelity [211,212]. In semantic communication, pedagogical and eclectic methods are particularly attractive, as they can be applied across heterogeneous architectures without modifying the underlying encoder-decoder pipeline.

From a computational standpoint, rule extraction typically involves fitting interpretable surrogate models such as decision trees or sparse rule sets, with runtime complexity dependent on the number of sampled semantic instances and extracted rules. Memory overhead is dominated by storage of the rule set itself, which remains modest compared to gradient-based or sampling-heavy explanation techniques. This makes rule extraction suitable for deployment on edge devices or control-plane components where interpretability is required with limited computational resources.

Use Case Example: In an AI-powered customer service application employing semantic communication, rule extraction can explain why a compressed semantic query is routed to a specific response module. For example, a rule such as “*IF topic = billing AND sentiment = negative, THEN route to escalation unit*” makes the system's decision logic explicit and auditable. Such explanations enable domain experts to validate routing policies and adjust semantic thresholds without directly modifying the underlying neural model.

Despite these advantages, rule extraction techniques have important limitations. The extracted rule set may fail to fully capture complex non-linear interactions present in deep semantic models, leading to reduced fidelity. Rule quality can also degrade under distribution shifts or when semantic representations are highly entangled. Furthermore, large rule sets may become difficult to interpret, undermining their original purpose. Consequently, rule extraction is most effective when used as a complementary explainability mechanism, providing high-level logical insight alongside feature-based and visualization-driven explanations in XAI-enabled semantic communication systems.

6. Counterfactual Explanations: Counterfactual explanations aim to explain model decisions by identifying alternative input scenarios that would lead to a different outcome. Rather than describing why a prediction was made, counterfactuals address *what would need to change* in the input to alter the model's decision. In semantic communication systems, where compressed semantic representations convey meaning rather than raw data, counterfactual explanations are particularly effective at revealing how small semantic modifications influence downstream interpretation or classification [213].

Formally, let $f_\theta : S \rightarrow \mathcal{Y}$ denote a semantic communication model mapping semantic representations $s \in S$ to outputs $y \in \mathcal{Y}$. Given an instance s with prediction $y = f_\theta(s)$, a counterfactual explanation seeks an alternative semantic representation s' such that

$$f_\theta(s') \neq f_\theta(s) \quad (25)$$

while minimizing a distance measure $d(s, s')$ and satisfying semantic plausibility constraints. This formulation ensures that the generated counterfactual corresponds to a minimal and meaningful semantic change rather than an arbitrary perturbation.

In semantic communication, counterfactuals operate naturally at the semantic representation and decision layers, where features correspond

to interpretable tokens, concepts, or objects. Generating counterfactual explanations typically involves optimization-based search or generative modeling, with computational cost depending on the dimensionality of the semantic space and the complexity of the plausibility constraints. While such methods may be computationally intensive for large or highly entangled representations, lightweight approximations or constrained searches can make counterfactual generation feasible on edge devices for selected instances, with more complex analyses offloaded to edge servers or cloud infrastructure.

Use Case Example: Consider a semantic communication model used in a smart healthcare assistant that classifies transmitted messages as *urgent* or *routine*. A counterfactual explanation may indicate that replacing the semantic token “chest pain” with “fatigue” changes the predicted class from urgent to routine. This directly exposes which semantic elements drive high-stakes decisions and allows clinicians to assess whether the model's behavior aligns with medical reasoning.

Despite their interpretability benefits, counterfactual explanations also present limitations. Generating plausible and realistic counterfactuals can be challenging in high dimensional semantic spaces, and poorly defined distance metrics may lead to semantically invalid alternatives. In addition, counterfactuals may exploit spurious correlations in the model, potentially misleading users about true causal relationships. As a result, counterfactual explanations are most effective when combined with complementary explanation methods, such as feature importance or rule-based techniques, to provide a more robust and trustworthy interpretability framework for XAI-enabled semantic communication systems.

7. Explainable Reinforcement Learning. Reinforcement Learning (RL) enables an agent to learn decision-making policies through interaction with an environment, guided by reward feedback. In semantic communication systems, RL is naturally suited for adaptive control at the semantic decision and resource management layers, where actions such as semantic compression level, transmission timing, or policy selection must be optimized under dynamic channel conditions and user requirements [214]. This adaptability is particularly valuable in interactive applications, including intelligent assistants, personalized translation, and context-aware semantic encoders.

Formally, an RL-based SemCom system can be modeled as a Markov Decision Process (MDP) defined by the tuple $(S, \mathcal{A}, P, R, \gamma)$, where S denotes the semantic state space (e.g., semantic embeddings, channel quality indicators, user context), $\mathcal{A}$ the action space (e.g., compression strategy, transmission mode), P the state transition dynamics, R the reward function, and γ the discount factor. The agent learns a policy $\pi(a | s)$ that maximizes the expected cumulative reward. Explainability in this setting focuses on interpreting why a particular action a was selected in a given semantic state s , and how different reward components influence the learned policy.

Several explainability mechanisms have been proposed for RL. Saliency and attribution methods can be applied to policy networks to identify which semantic state features most influence action selection at each decision step. Reward decomposition techniques partition the overall reward into interpretable components, such as semantic fidelity, latency, or energy efficiency, thereby clarifying the agent's optimization priorities [215]. In addition, policy summarization methods approximate the learned policy using interpretable surrogates, such as decision trees or rule sets, providing high-level insights into the agent's long-term behavior. Feature attribution approaches, including SHAP or LIME, can also be applied locally to quantify the contribution of specific semantic features or actions to observed outcomes [216].

From a computational perspective, incorporating explainability into RL introduces additional overhead beyond standard policy inference. Saliency-based explanations typically require gradient computation with respect to policy outputs, while reward decomposition and policy summarization involve auxiliary models or post-processing steps. These costs scale with the dimensionality of the semantic state and ac-

Table 10
Advanced XAI methods and Python libraries suitable for semantic communications.

References	Category	XAI Methods	Python Libraries
[177–179]	Feature Attribution	Feature Importance	SHAP, LIME, ELI5, interpret, Dalex, AI Explainability 360, Captum, Skater, SHAPash, PyCEbox, Treeinterpreter, DeepLIFT, WeightWatcher
[180–182]	Instance-Based	Subset of Features Guaranteeing a Prediction	DiCE, Anchor, LimeTabular, AnchorTabular, certifi, PDPbox
[183–185]	Self-Explaining	Models that are Explainable by Design	interpretML, ExplainX.ai, TrustyAI, RAI, Concept Activation Vectors, Alibi Detect
[186–190]	Uncertainty Estimation	Quantify Reliability of Prediction	Fairlearn, Fairness Indicators, Fairness, Fairness Comparison, certifi, PrivacyRaven
[191–194]	Counterfactual	Alternative Scenarios	DiCE, Alibi, Counterfactuals, H2O.ai Explainability
[100,115,195,196]	Human-Readable Rule Extraction	Extract Human-Readable Rules	ELI5, interpret, AI Explainability 360, Treeinterpreter, RAI
[116,197,198]	Attention Mechanisms	Highlight Important Features or Regions	Captum, Grad-CAM, Attention Networks
[100,199,200]	NLP Techniques	Generate Explanations in Human-Readable Language	AI Explainability 360, interpretML, ELI5, SHAPash
[201,202]	Model Confidence	Conveying Model Confidence and Uncertainty	Fairlearn, Fairness Indicators, FAT Forensics, TrustyAI
[203–205]	Adversarial Explanation	Testing with Intentionally Modified Inputs	CleverHans, Adversarial Robustness Toolbox (ART), PrivacyRaven, Fairkit-learn
[188,201,206,207]	Interactive Explanations	Actively Engage and Explore Decision Pathways	What-If Tool, certifi, Fairlearn, TrustyAI
[206,208–210]	Human Feedback Loops	Incorporating Iterative Feedback Loops	TrustyAI, What-If Tool, Fairlearn

tion spaces and may increase memory usage due to storage of trajectories or intermediate activations. In practice, lightweight explainability techniques can be applied selectively on edge devices, whereas more detailed policy analysis and summarization are better suited for edge servers or cloud platforms.

Use Case Example: In a task-oriented semantic communication system for smart assistants, an RL agent may dynamically decide whether to transmit a full semantic representation or a compressed symbolic message based on channel quality and task urgency. By applying reward decomposition and feature attribution, system designers can inspect whether the agent prioritizes semantic accuracy, latency, or energy efficiency, and adjust the reward design accordingly. This transparency is essential for validating agent behavior in real-world scenarios involving competing performance objectives.

Despite their advantages, explainable RL approaches face important challenges. Explanations may be unstable across training iterations, sensitive to stochastic exploration, or difficult to interpret in high-dimensional semantic state spaces. Moreover, surrogate explanations may not fully capture long-term dependencies encoded in the policy, potentially leading to misleading interpretations. Consequently, explainable RL techniques should be carefully designed and combined with complementary explanation methods to ensure reliable and trustworthy semantic communication systems.

8. Natural Language Generation (NLG). Natural Language Generation (NLG) enables AI systems to produce human-readable textual explanations that describe internal decisions, reasoning processes, or outcomes. In semantic communication systems, NLG is particularly well suited for the explanation and interaction layers, where the goal is to convey not only *what* semantic content was transmitted, but also *why* specific semantic decisions were made. By translating internal representations and explanation signals into natural language, NLG provides an intuitive interface between complex AI-driven communication pipelines and human users.

Formally, let $f_\theta : S \rightarrow \mathcal{Y}$ denote a semantic communication model and let $E(s)$ represent an intermediate explanation signal derived from the model (e.g., feature attributions, rules, or counterfactuals). An NLG-based explainer can be modeled as a conditional language generator g_ϕ that maps explanation inputs to textual descriptions,

$$g_\phi : (S, E) \rightarrow \mathcal{T} \quad (26)$$

where $\mathcal{T}$ denotes the space of natural language explanations. This formulation highlights the role of NLG as an explanation delivery mechanism that summarizes and contextualizes outputs from other XAI techniques, rather than directly attributing causality at the feature level.

In practice, NLG models are trained or fine-tuned to generate explanations aligned with domain-specific semantics, enabling them to verbalize decisions such as semantic prioritization, compression choices, or transmission policies. From a computational perspective, NLG introduces overhead associated with sequence generation, with runtime and memory costs scaling with model size and output length. Lightweight template-based or small language models may be deployed on edge devices for real-time explanation generation, whereas more expressive neural generators are typically executed on edge servers or cloud infrastructure.

Use Case Example: In a medical semantic communication system, an AI assistant may prioritize the transmission of high-risk clinical information under bandwidth constraints. An NLG module can generate an explanation such as: “*The system prioritized recent abnormal ECG patterns due to elevated cardiac risk, while omitting stable historical vitals to reduce transmission load.*” Such explanations allow clinicians to understand both the transmitted semantic content and the reasoning behind selective communication.

Despite their advantages, NLG-based explanations present important challenges. Generated text may suffer from hallucination or overgeneralization, potentially misrepresenting the underlying model behavior. Ensuring faithfulness between textual explanations and actual semantic decisions remains a critical concern, particularly when NLG models are loosely coupled with the decision process. Moreover, generating verbose explanations may increase latency or bandwidth usage if explanations themselves are transmitted. As a result, NLG is most effective when used in conjunction with grounded explanation signals and appropriate constraints to ensure accuracy, conciseness, and trustworthiness. Overall, NLG enhances explainability in semantic communication by transforming complex semantic decisions into accessible, human-aligned narratives, thereby improving transparency, usability, and user trust in AI-enabled communication systems [217,218].

In summary, the integration of XAI techniques into semantic communication systems represents a pivotal advancement toward building transparent, reliable, and human aligned communication frameworks. Through a detailed analysis of rule-based explanations, feature attribution methods like SHAP and LIME, visualization approaches such as

Layer-wise Relevance Propagation (LRP), and symbolic techniques like rule extraction and counterfactual reasoning, we observe a growing emphasis on interpretability across diverse SemCom architectures. Each method contributes uniquely: rule-based models clarify decision logic, feature attribution reveals semantic relevance, visualizations enhance transparency in perception tasks, and counterfactuals aid in exploring alternative scenarios for robust reasoning. Additionally, emerging techniques such as the use of knowledge graphs and ontologies enable structured semantic understanding and reasoning, while reinforcement learning frameworks equipped with XAI layers support adaptive and explainable policy optimization. Natural Language Generation further enriches user experience by translating model behavior into comprehensible text, making AI-driven communication more accessible. The Table 10 complements this discussion by providing a structured mapping between cutting edge XAI methods and the Python libraries that support their practical implementation. This resource can empower both researchers and system designers to integrate suitable explainability modules tailored to specific semantic communication use cases. Altogether, this section underscores that the convergence of XAI with semantic communication is not merely an add-on but a fundamental requirement for deploying trustworthy and user-centric AI systems. As the field matures, future research should focus on hybrid and multimodal explainability, interactive feedback mechanisms, and domain-specific tailoring of explanation models to meet the nuanced demands of evolving communication technologies.

5. Open issues, potential solutions and future research directions in XAI-enabled semantic communication

AI-enabled semantic communication is redefining the foundations of modern wireless systems by shifting the communication paradigm from the reliable transmission of raw symbols to the exchange of contextually meaningful information. While recent advances in deep learning have substantially improved the capability of semantic encoders and decoders, their increasing complexity introduces new challenges related to transparency, accountability, robustness, and human interpretability. As semantic communication systems become more deeply embedded in safety-critical, real-time, and large-scale applications, the availability of explainable and trustworthy artificial intelligence (XAI) mechanisms becomes not merely advantageous, but essential.

To systematically examine these challenges, we organize the open research issues and potential solution pathways along two primary dimensions: **Explainability and Interpretability**, and **Interoperability and Standardization**. In this context, security and privacy risks that arise from explainability mechanisms (e.g., explanation leakage and manipulation) are treated as integral subtopics within the explainability and interpretability dimension. This categorization provides a structured and actionable framework for guiding future research toward semantic communication systems that are not only intelligent and efficient, but also transparent, robust, and aligned with human understanding.

Fig. 11 illustrates a conceptual overview of the major open challenges and associated research directions, which serve as the basis for the discussion in the following subsections.

5.1. Security risks introduced by explainability in semantic communication

Explainable AI (XAI) has emerged as a foundational component for building transparent, trustworthy, and human-centric semantic communication systems. By revealing how semantic encoders, decoders, and adaptive transmission strategies operate, explainability supports system validation, debugging, and user trust. However, alongside these benefits, explainability introduces a subtle yet critical class of security and privacy risks that are not adequately captured by traditional communication security models.

Unlike conventional threats that primarily target the physical channel, transmitted symbols, or semantic payloads, explainable outputs themselves can act as an unintended information leakage channel. Attribution maps, saliency scores, counterfactual explanations, and confidence indicators may expose fine-grained semantic dependencies, latent feature importance, or model sensitivities that were never intended for external disclosure. When such information is accessible to adversaries either through direct interaction with the explanation interface or via intercepted explanation transmissions it can be exploited to infer sensitive attributes, reconstruct aspects of the training data, or gain insights into the internal behavior of the semantic model [219].

These risks are particularly pronounced in semantic communication systems due to the tight coupling between semantic representations, channel-aware adaptation, and task-oriented decision making. Explanations may remain informative even when the semantic message itself is encrypted or otherwise protected, effectively creating a semantic side-channel that bypasses conventional security mechanisms. Moreover, explainable outputs are not only vulnerable to passive information leakage but may also be actively manipulated. Carefully crafted perturbations at the input or channel level can significantly alter explanation patterns while leaving the reconstructed semantic output largely intact, potentially misleading users or downstream decision processes. This observation highlights that semantic correctness alone is insufficient to guarantee trustworthy system behavior when explanations are involved.

5.1.1. Adversarial model and security assumptions.

In line with the security challenges summarized in Table 11, we consider a realistic and moderately strong adversarial model tailored to XAI-enabled semantic communication systems. The protected assets include the original and reconstructed semantic messages ($M, \hat{M}$), latent semantic representations (s, z), explanation outputs E (e.g., saliency maps, attribution vectors, or counterfactual descriptions), and user feedback signals used for online adaptation. The adversary is assumed to have access to the explanation interface or explanation payloads, may eavesdrop on explanation transmissions, and may inject bounded perturbations at the input or channel level. In addition, the adversary may correspond to an honest-but-curious edge server or a co-located tenant in a multi-tenant deployment.

Under this model, the adversarial objectives include inferring sensitive attributes or training membership through explanation leakage, manipulating explanations to distort human interpretation while preserving semantic output quality, and extracting information about model behavior through repeated explanation queries. The corresponding security objective is therefore to preserve semantic fidelity and system utility while constraining the information leakage and manipulation risks introduced by explainable outputs. This formulation directly motivates the mitigation strategies discussed earlier, including explanation sanitization, privacy-preserving explanation generation, and controlled access to explanation interfaces.

- **Illustrative Vulnerability: Attribution Manipulation under Channel Perturbations.**

To illustrate the vulnerabilities introduced by explainability, consider a semantic communication system equipped with gradient-based or attention-based explanation mechanisms, such as Integrated Gradients or Grad-CAM. An adversary applies a bounded perturbation δ to the input or channel such that the reconstructed semantic message remains semantically consistent with the original, while the generated explanation undergoes substantial change. This behavior can be conceptualized through the following constrained optimization problem:

$$\max_{\delta \in S} D(E(x), E(x; \delta)) \quad \text{s.t.} \quad F_{\text{sem}}(x, \hat{x}) \geq \tau, \quad (27)$$

where $D(\cdot)$ denotes a distance measure between explanations, S represents a bounded perturbation set, and τ is a semantic fidelity threshold.

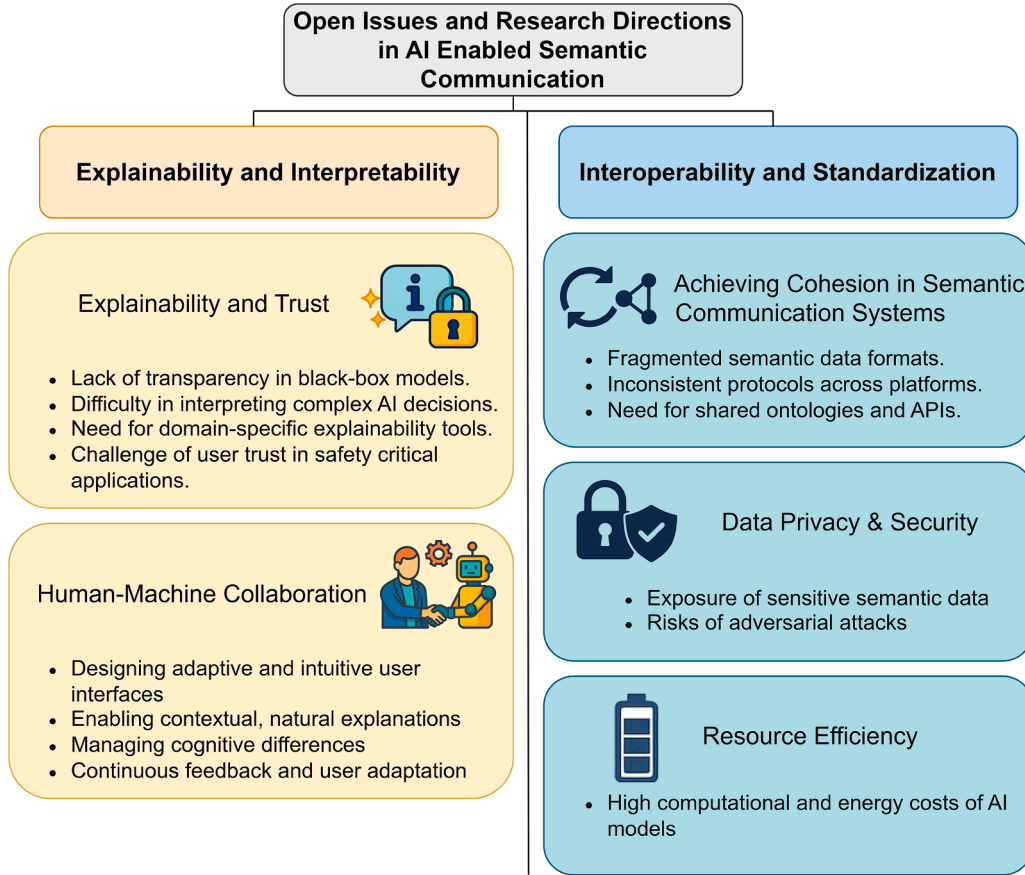


Fig. 11. Open issues and challenges in AI-enabled semantic communication.

Table 11

Mitigation strategies against explanation leakage and manipulation in XAI-enabled semantic communication systems.

Mitigation	Threat	Integration	Overhead
Explanation sanitization (top-K, clipping, quantization)	Membership / attribute inference	XAI engine / interface	Low compute; reduced payload
Differential privacy for explanations	Training data leakage	XAI engine	Noise; negligible bandwidth
Robust or smoothed explanations	Attribution manipulation	XAI engine	Higher compute (η_{kai})
Encryption and authentication of explanations	Eavesdropping / tampering	Protocol layer	Small crypto overhead
Access control and rate limiting	Model extraction	Interface / control plane	Minor latency

In practice, such perturbations can preserve task performance and semantic similarity while significantly degrading explanation stability, effectively redistributing or flipping attribution patterns. As a result, explanations may no longer faithfully reflect the underlying semantic reasoning of the system, even though the observable semantic output appears correct. This example underscores that explanation robustness constitutes a distinct and essential security requirement, complementing traditional notions of semantic fidelity and communication reliability.

5.2. Explainability and interpretability

In AI-enabled semantic communication, ensuring explainability and interpretability is crucial for the reliability and transparency of the system. This sub-section delves into the major challenges associated with achieving explainability and interpretability and explores possible solutions to address these issues. By enhancing our understanding of the decision making processes within these systems, we aim to build more trustworthy and effective communication technologies.

1. Challenge: Explainability and Trust

Ensuring that AI-driven decisions are both interpretable and trustworthy is essential for fostering user confidence in AI-enabled semantic communication systems. These systems, which rely on transmitting seman-

tically meaningful information rather than raw data, often incorporate complex AI models whose decision making processes are not inherently transparent. This opacity can hinder user understanding, impede model validation, and ultimately limit real world adoption. Therefore, there is a pressing need for advanced explainability techniques tailored to the semantic nature of these systems and aligned with user expectations [159].

Potential Solutions:

• Domain-Specific and Interactive Explainability Techniques:

To enhance interpretability in SemCom, explainability methods must be adapted to the specific characteristics of the communication modality. For example, natural language generation techniques are well suited for textual SemCom, while saliency maps and attention heatmaps provide intuitive insights in visual and multimodal settings [220]. Interactive tools that enable users to inspect model decisions at various levels of granularity such as through stepwise decision tracing or feature relevance overlays can significantly improve user comprehension. Such interactivity not only demystifies how semantic information is encoded, transformed, and decoded but also allows users to actively probe and verify AI behavior. This interactivity is critical for real-time and high stakes SemCom applications, where decision accountability is vital [221].

- **Model-Agnostic Approaches and User-Centric Design:**

Model-agnostic XAI techniques such as LIME and SHAP [99] are instrumental in offering transparent, interpretable outputs for black-box AI models, which are prevalent in semantic encoders and decoders. These methods attribute importance to individual input features words, symbols, or frames clarifying their contribution to a given output. Integrating these techniques into SemCom pipelines enables explanation generation without altering model architectures, thus preserving performance while improving transparency. Additionally, employing user-centric design principles ensures that these explanations are tailored to the cognitive needs of diverse users. For instance, adaptive explanation interfaces that adjust explanation complexity based on user expertise can bridge the gap between AI developers, domain experts, and end users. Continuous refinement through user feedback and iterative design cycles further ensures that explanation mechanisms remain relevant, effective, and aligned with evolving user expectations [221].

2. Challenge: Human-Machine Collaboration

Effective human-machine collaboration in AI-enabled semantic communication systems demands a balance between technological sophistication and user accessibility. One of the primary challenges lies in designing intuitive and adaptive interfaces that accommodate users with diverse backgrounds and varying levels of technical expertise [222]. These interfaces must be context-aware, dynamically adjusting to domain-specific applications such as healthcare, customer service, or industrial automation [21]. Another major challenge is enabling AI systems to deliver assistance that is both timely and semantically relevant, without overloading the user. This requires robust NLP capabilities that can interpret and respond in human understandable formats, thereby enhancing transparency and interpretability [223]. Bridging the cognitive gap between human intuition and machine reasoning also remains a significant barrier to collaboration. Moreover, maintaining alignment with user expectations necessitates mechanisms for continuous learning and system adaptation based on real-time feedback.

Potential Solutions:

- **Enhancing Interfaces and Communication Strategies:**

To improve collaboration, SemCom systems must prioritize user-centric interface design. Interfaces should be adaptable across application domains and customizable based on user expertise. Employing modular and context-aware design strategies allows these systems to dynamically tailor interactions, ensuring clarity and usability [21]. Furthermore, integrating advanced NLP and dialogue generation models can enable AI agents to deliver contextually appropriate, explainable feedback. Visual and interactive explanation components such as natural language explanations, saliency maps, or decision flow diagrams can further bridge the gap between opaque model behavior and user understanding [223]. Interactive tools that allow users to query the AI system regarding specific decisions help foster transparency and improve trust in the collaboration process.

- **Bridging Cognitive Gaps and Enabling Continuous Feedback Loops:**

Addressing the cognitive mismatch between humans and AI requires simplifying the interpretability of complex model processes. Visualization techniques that translate high dimensional internal states into comprehensible diagrams can support this goal. For example, LRP or attention based heatmaps can visually represent how specific semantic features influence model outputs. Additionally, establishing real-time feedback mechanisms where user reactions, corrections, or queries directly influence model updates creates a loop of collaborative learning and adaptation [222]. These adaptive systems should support continuous updates informed by recent user interactions and domain specific refinements, thereby sustaining relevance and reliability in evolving communication environments.

5.3. Interoperability and standardization

In AI-enabled semantic communication systems, ensuring seamless interoperability across heterogeneous platforms while adhering to evolving standards is essential for achieving scalable and transparent communication infrastructures. The fusion of AI technologies with communication frameworks introduces unique complexities especially when diverse devices, networks, and vendors must share semantically rich data. Without harmonized protocols and standardized semantic data representations, inconsistencies in how meaning is encoded, transmitted, and interpreted can hinder the performance, trustworthiness, and explainability of semantic models [224,225].

A particular challenge lies in ensuring that AI-generated content whether in text, speech, or multimodal form maintains consistent semantic fidelity across systems. Since XAI techniques often rely on understanding how semantic elements influence model behavior, the lack of standardized formats and interfaces can disrupt the interpretability of outputs. This makes the pursuit of standardization not merely an engineering challenge, but a foundational requirement for enabling explainable and reliable semantic communication.

1. Challenge: Achieving Cohesion in AI-Enabled Semantic Communication Systems:

To facilitate effective and explainable AI-driven communication, it is imperative that systems adopt a shared semantic understanding of transmitted content. Without common ground in how meaning is represented and exchanged, XAI tools such as LIME, SHAP, or visualization based interpretability methods struggle to consistently map model outputs back to semantically meaningful elements. Thus, developing unified semantic ontologies, machine interpretable formats, and interoperable application programming interfaces (APIs) is crucial for enabling explainable cross-domain interaction [226].

Potential Solutions:

- **Standardized Semantic Data Formats:**

Adopting universally accepted data representations, such as Resource Description Framework (RDF) and JavaScript Object Notation for Linked Data (JSON-LD), allows semantic communication systems to represent meaning consistently across devices and vendors. These formats enable downstream XAI techniques to interpret inputs and outputs uniformly, enhancing traceability and transparency. For instance, metadata schemas and ontologies can be extended with explainability annotations, helping models and human users track how semantic elements are derived, compressed, or filtered in the communication process [227,228]. This forms the basis for explanation models to connect specific features to their role in semantic decision-making.

- **Protocol-Level Support for Explainability:**

Beyond standardizing content, communication protocols must also be designed to preserve semantic explainability. This includes supporting data serialization and compression strategies that retain meaningful structure for post-hoc analysis. Lightweight, explainability aware protocols potentially adapted from Message Queuing Telemetry Transport (MQTT) or Constrained Application Protocol (CoAP) should accommodate semantic metadata and reasoning traces to support downstream interpretation tools. For instance, integrating SPARQL Protocol and RDF Query Language (SPARQL) endpoints into communication workflows allows explainable querying of transmitted content, supporting real-time inspection and explanation of AI decisions across distributed networks [52,229,230]. Secure and modular protocol designs ensure that semantic insights remain interpretable while safeguarding data integrity.

2. Challenge: Data Privacy and Security: The integration of AI into semantic communication systems involves the processing of large volumes of data, much of which is sensitive and critical to users, organizations, and infrastructure. Ensuring strong data privacy and robust security is essential not only for legal and ethical compliance but also

for fostering trust and preventing adversarial threats [47]. In parallel, as semantic communication systems increasingly rely on complex AI models to understand and transmit meaning, the need for explainability becomes fundamental. Users must not only be assured that their data is protected but also understand how decisions regarding semantic encoding, compression, or transmission are made. Integrating XAI thus serves a dual purpose enhancing transparency and contributing to the overall robustness and accountability of these systems [231].

Potential Solutions:

- **Privacy-Preserving AI:** Ensuring the confidentiality of semantic data while enabling effective model training has led to the adoption of privacy preserving machine learning techniques. Federated Learning allows AI models to be trained locally across decentralized devices, ensuring that raw user data never leaves the device, thereby safeguarding privacy [232]. Homomorphic Encryption enables computations to be performed directly on encrypted data, eliminating the need to decrypt sensitive information during inference [233]. Differential Privacy introduces noise into the data to prevent identification of individual records while preserving aggregate patterns [234]. These techniques not only ensure compliance with data protection standards but also support semantic fidelity by allowing safe access to critical patterns without compromising the underlying semantics. Furthermore, integrating explainability techniques within these privacy preserving frameworks can help elucidate how protected features influence decisions without revealing the data itself, making AI-driven semantic communication both secure and interpretable.
- **Robust Security Mechanisms:** Securing AI-enabled semantic communication requires a layered approach combining encryption, authentication, and anomaly detection. Advanced encryption standards must be implemented for both data in transit and data at rest. Identity verification protocols ensure only authorized users and devices can engage in semantic exchanges. Additionally, Intrusion Detection Systems powered by AI can detect unusual semantic patterns that may indicate malicious activity [21]. To mitigate adversarial attacks where inputs are intentionally perturbed to manipulate AI predictions. Adversarial Training can be employed to expose models to both clean and malicious data during training, enhancing resilience [235]. The inclusion of XAI techniques, such as LRP, further allows inspection of how models interpret semantic features under threat, enabling system designers to pinpoint vulnerabilities and enhance defensive mechanisms. This synergistic integration of security and explainability ensures that semantic communication remains trustworthy, resilient, and accountable even under adversarial conditions [236].

3. Challenge: Resource Efficiency: While AI models have demonstrated exceptional capabilities across various domains, they often come with significant computational and energy requirements. This creates considerable barriers for deploying such models in resource constrained environments, particularly in IoT and edge devices that support AI-enabled semantic communication systems. In these settings, maintaining semantic fidelity and interpretability without exceeding energy or memory constraints is a critical challenge. The trade-off between model accuracy, inference time, and resource utilization becomes especially pronounced in scenarios where devices operate on limited battery power, minimal storage, or constrained processing capabilities [237,238]. To facilitate practical implementation and scalability, semantic communication systems must adopt resource efficient AI techniques that retain high performance while minimizing computational burden [224].

Potential Solutions:

- **Model Compression for Semantic AI:** To reduce the overhead associated with deploying AI models in low resource environments, model compression techniques such as pruning, quantization, and knowledge distillation are essential. Model pruning simplifies neural networks

by removing non-critical weights or neurons, resulting in a leaner architecture with reduced computational demands [239]. Quantization reduces the bit width of numerical values used in model computations, decreasing memory usage and accelerating inference time, especially on specialized hardware [240]. Knowledge distillation transfers the learned behavior of a large, complex model (teacher) to a smaller, more efficient model (student), preserving semantic reasoning capabilities while reducing the resource footprint [241]. These techniques are instrumental in enabling semantic communication models to maintain performance even on devices with limited storage, bandwidth, and compute capacity. Moreover, when combined with explainability frameworks, lightweight models can still offer transparent and interpretable insights, maintaining the trustworthiness and accountability of semantic decisions.

- **Energy-Efficient AI Algorithms and Hardware:** Beyond software-level optimizations, energy efficient algorithm design and specialized hardware are vital for sustainable AI-enabled semantic communication. Algorithms must be optimized for low power inference without sacrificing semantic accuracy an objective often achieved using adaptive neural architectures or event driven processing models [242]. Simultaneously, the development of AI-specific chips with low power cores, energy aware scheduling, and on chip explainability support (e.g., saliency mapping) can help bring real-time interpretable semantic reasoning to the edge [243,244]. These advancements not only extend battery life and reduce operational costs but also ensure semantic communication systems remain deployable in remote, mobile, or intermittently connected environments.

In conclusion, achieving resource efficiency in AI-enabled semantic communication systems is not merely a technical optimization it's a foundational requirement for widespread, real world deployment. As semantic communication shifts toward ubiquitous, low power platforms such as IoT and edge devices, the integration of model compression strategies and energy aware AI becomes increasingly vital. Techniques like pruning, quantization, and knowledge distillation ensure that performance and interpretability can be preserved without overwhelming hardware limitations. Meanwhile, advancements in energy efficient algorithm design and low power hardware architectures further reinforce the feasibility of deploying semantic AI in constrained environments. Crucially, these solutions must work in harmony with explainability mechanisms to ensure that compact, efficient models remain transparent, trustworthy, and user aligned. Together, these innovations form a sustainable path forward, enabling intelligent, interpretable, and resource aware communication systems capable of thriving across diverse and demanding applications.

6. Conclusion

The evolution toward 6G and beyond is transforming communication systems from bit-centric pipelines into intelligent infrastructures that reason about and transmit semantic meaning. While artificial intelligence enables this shift, it also introduces opacity through complex black-box models, raising concerns about transparency, reliability, and trust. To address this challenge, this paper presented a focused and systematic survey on the integration of Explainable Artificial Intelligence (XAI) within semantic communication systems.

We revisited the foundations of semantic communication and proposed a modular XAI-enabled architecture that embeds explainability across the entire communication pipeline, from semantic source encoding to user-in-the-loop adaptation. A unified set of performance metrics was introduced to jointly evaluate semantic fidelity, efficiency, and explanation quality. Our review reveals that, despite significant progress in semantic communication, explainability remains largely underexplored and is typically treated as an afterthought rather than a design principle. To bridge this gap, we introduced a fine-grained taxonomy of opportunities that maps semantic communication layers

and sub-problems to suitable XAI paradigms, providing practical guidance for future system design.

In summary, the convergence of XAI and semantic communication enables a new class of trust-aware intelligent networks that are efficient, interpretable, and aligned with human expectations. This work is expected to stimulate further research toward building transparent and robust semantic communication systems capable of supporting safety-critical applications in healthcare, autonomous systems, and industrial automation.

CRedit authorship contribution statement

Muhammad Furqan Zia: Writing – review & editing, Writing – original draft, Visualization, Methodology, Investigation, Formal analysis, Data curation, Conceptualization; **Messaoud Ahmed Ouameur:** Writing – review & editing, Validation, Project administration; **Miloud Bagaa:** Writing – review & editing; **Daniel Massicotte:** Writing – review & editing; **Adlen Ksentini:** Writing – review & editing.

Data availability

No data was used for the research described in the article.

Declaration of competing interest and the use of Generative AI-Assisted Technologies in the Writing Process

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

During the preparation of this survey, generative AI tools (e.g., ChatGPT by OpenAI) were used to support drafting, formatting, and language refinement. All technical content, conceptual framing, and analytical insights were developed and validated solely by the authors. The authors accept full responsibility for the integrity and accuracy of the final manuscript.

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